

## Article

# Multidimensional Evaluation Framework and Classification Strategy for Low-Carbon Technologies in Office Buildings

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## Abstract

The global climate crisis has driven unprecedented agreements among nations on carbon mitigation. With China's commitment to carbon peaking and carbon neutrality targets, the building sector has emerged as a critical focus for emission reduction, particularly because office buildings account for over 30% of building energy consumption. However, a systematic and regionally adaptive low-carbon technology evaluation framework is lacking. To address this gap, this study develops a multidimensional decision-making system to quantify and rank low-carbon technologies for office buildings in Beijing. The method includes four core components: (1) establishing three archetypal models—low-rise ( $H \leq 24$  m), mid-rise ( $24 \text{ m} < H \leq 50$  m), and high-rise ( $50 \text{ m} < H \leq 100$  m) office buildings—based on 99 office buildings in Beijing; (2) classifying 19 key technologies into three clusters—Envelope Structure Optimization, Equipment Efficiency Enhancement, and Renewable Energy Utilization—using bibliometric analysis and policy norm screening; (3) developing a four-dimensional evaluation framework encompassing Carbon Reduction Degree (CRD), Economic Viability Degree (EVD), Technical Applicability Degree (TAD), and Carbon Intensity Degree (CID); and (4) conducting a comprehensive quantitative evaluation using the AHP-entropy-TOPSIS algorithm. The results indicate distinct priority patterns across the building types: low-rise buildings prioritize roof-mounted photovoltaic (PV) systems, LED lighting, and thermal-break aluminum frames with low-E double-glazed laminated glass. Mid- and high-rise buildings emphasize integrated PV-LED-T8 lighting solutions and optimized building envelope structures. Ranking analysis further highlights LED lighting, T8 high-efficiency fluorescent lamps, and rooftop PV systems as the top-recommended technologies for Beijing. Additionally, four policy recommendations are proposed to facilitate the large-scale implementation of the program. This study presents a holistic technical integration strategy that simultaneously enhances the technological performance, economic viability, and carbon reduction outcomes of architectural design and renovation. It also establishes a replicable decision-support framework for decarbonizing office and public buildings in cities, thereby supporting China's "dual carbon" goals and contributing to global carbon mitigation efforts in the building sector.

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**Keywords:** carbon emissions; low-carbon technologies; evaluation system; multidimensional evaluation; office buildings

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## 1. Introduction

The global climate crisis has catalyzed an unprecedented level of international consensus on carbon mitigation, with over 130 nations committing to carbon neutrality targets under the Paris Agreement [1]. As the world's largest carbon emitter, China faces significant challenges in balancing economic growth with its emission reduction commitments. China has officially committed to achieving a carbon peak by 2030 and carbon neutrality by 2060—a transition that requires profound transformations across all sectors of the economy and society. The building sector has been identified as one of the top ten prioritized domains for emission control [2].

In this context, the building sector plays a pivotal role in global mitigation efforts [3,4]. Statistics show that emissions from the entire construction process—design, construction, and operation stages [5,6]—account for 38% of global greenhouse gas emissions [7], while this proportion rises to 50.6% in China [8]. Among various building types, office buildings are critical targets for carbon reduction owing to their high occupancy rates, long-term operations, and substantial energy demands. According to local statistics in China's megacities, office buildings consume over 30% of the total energy consumption of buildings. The energy consumption per unit area of office buildings is five to eight times that of ordinary residential buildings [9]. Notably, numerous office buildings exhibit relatively low energy efficiency [10], leading to significant carbon emissions. Therefore, office buildings have become a key focus for achieving carbon neutrality in the construction industry.

The development and utilization of low-carbon technologies have garnered significant attention during the decarbonization of the building sector [11,12]. Recent advancements in building low-carbon technologies have revealed multidimensional innovation pathways, including the selection of sustainable building materials [13], improvements in the energy efficiency of building equipment [14], optimization of envelope structures [15], enhancement of architectural design [16], and integration of renewable energy systems [17].

Extending building lifespans and substituting conventional materials with low-carbon alternatives have already become increasingly prevalent practices [18]. High-efficiency equipment combined with intelligent control systems has been demonstrated to reduce operational energy consumption by 20–40% [19]. Envelope optimization—achieved through advanced insulation and dynamic shading—further suppresses peak cooling loads [20]. Feng et al. (2019) demonstrated that passive strategies such as daylighting and natural ventilation can reduce annual cooling demand by 18–35% across 34 net-zero energy building cases in hot climates, effectively complementing active systems [21]. In the context of renewable energy utilization, De Wolf et al. (2017) pointed out that the application of technologies such as solar energy, wind energy, and ground source heat pumps can lead to an exponential decrease in carbon emissions from building operations, with benefits increasing as technologies mature [2]. Most recently, Fang et al. (2025) highlighted that nano-confined catalysis, as a transformative strategy, can address the challenges of energy transition by precisely regulating the catalytic microenvironment, thereby significantly enhancing the potential for renewable energy utilization in the built environment [22].

However, achieving significant improvements in building energy conservation and carbon reduction cannot rely solely on individual energy-saving technologies. Instead,

integrated technical solutions that balance various low-carbon technologies are essential [23,24]. Empirical studies have shown that combining multiple low-carbon technologies can enhance energy efficiency in renovation projects by more than 16% [25] and reduce lifecycle carbon emissions by 22–41% [26,27].

Nevertheless, most existing research narrowly focuses on technical solutions with the highest carbon reduction potential, using simple evaluation indicators [28,29] such as the carbon emission index, annual per-unit-area carbon emissions [30], and carbon intensity [31]. In practice, selecting appropriate low-carbon building technologies is a complex process [32,33] influenced by coupled factors, including building characteristics (e.g., type, age, and spatial layout), environmental conditions (e.g., climate zone and grid carbon intensity), economic constraints (e.g., budget and cost), and technical feasibility (e.g., equipment compatibility and construction practicality) [34,35]. Therefore, scientifically assessing the emission reduction potential and comprehensive impact of different carbon reduction technologies—and determining the optimal technology combination through multidimensional integration—has become a key challenge for the industry's low-carbon transformation and sustainable development.

Scholars have made remarkable progress in the multidimensional evaluation of carbon reduction technologies for buildings. Most studies use the multi-criteria decision-making (MCDM) method to comprehensively consider multiple factors, support low-carbon building design [36,37], and evaluate strategies and tools for mitigating climate change [38]. By analyzing the required technologies and combining assessment dimensions such as environmental benefits [39,40], economic performance [41], and technical feasibility, scholars have evaluated technology combination schemes using MCDM [42–44].

Notable studies include Nygaard et al., who developed a four-stage optimization strategy for technology screening [45]. Liu et al. quantitatively evaluated 16 typical low-carbon renovation technologies using an environment-economy-society model [3]. Yu et al. proposed an optimization framework that integrates minimum building carbon emissions, the number of indoor adverse hours, and the global cost of buildings, thereby improving the rationality of selecting optimal technical solutions [46]. Panagiotidou et al. found that a cost-effective transformation plan aligned with existing market trends requires additional intervention measures, such as the adoption of high-efficiency heat pumps and photovoltaic (PV) systems, to achieve a carbon reduction of over 90% [47]. The empirical research by Galimshina, A. et al. further confirmed the importance of environmental and economic feasibility in technology selection [48].

To address the limitations of the systematicness and applicability of existing low-carbon technology evaluation systems, this study focuses on selecting low-carbon technologies for office buildings and proposes a four-dimensional MCDM evaluation method. By assessing key carbon reduction technologies and analyzing their impact weights, this study quantified the specific impacts of these technologies, providing scientific support for their architectural design and implementation.

The key innovations of this research are reflected in three aspects: (1) a four-dimensional evaluation system that incorporates four critical indicators—Carbon Reduction Degree (CRD), Economic Viability Degree (EVD), Technical Applicability Degree (TAD), and Carbon Intensity Degree (CID); (2) an improved Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) hybrid algorithm that integrates the Analytic Hierarchy Process (AHP) and entropy weighting to calculate combined weights, thereby enhancing the selection of optimal carbon reduction technology combinations for office buildings while considering diverse stakeholder preferences; and (3) prototype models developed

for three typical types of office buildings in Beijing, each featuring customized classification strategies. These models exhibit superior performance in terms of benefit assessment and strategy customization.

The theoretical and practical significance of this study lies in offering a comprehensive technical integration solution that simultaneously addresses technological effectiveness, economic feasibility, and carbon-reduction performance in architectural design and renovation. The research outcomes provide quantitative decision-making tools for selecting suitable technical solutions tailored to office buildings across various regions and types, ultimately contributing to the achievement of China's dual carbon goals in the construction sector.

## 2. Methods

### 2.1. Research Field and Data Source

As the capital of China, Beijing possesses a substantial stock of office buildings characterized by high energy consumption. The total floor area of office buildings in Beijing has reached 126 million square meters, accounting for 12.6% of the national total [49]. The carbon emission intensity per unit area is  $78.6 \text{ kgCO}_2/\text{m}^2\cdot\text{a}$ , which is 23% higher than the national average for public buildings. In 2022, CO<sub>2</sub> emissions from Beijing's office buildings amounted to 18.5 million tons, representing 42% of the city's building sector emissions [50]. Under the "dual carbon" goals, the "Beijing Construction Industry Development Plan for the 14th Five-Year Plan Period" explicitly sets a target to reduce office building carbon emissions by 23% by 2025 [51]. This underscores the urgent need for the low-carbon renovation of existing buildings and the adoption of near-zero-carbon technologies in new constructions. Beijing's office buildings offer significant research value as a key case study of low-carbon transformation in China's megacities.

Among the office building stock in Beijing, 64% were constructed between 2000 and 2022 [52], reflecting the technological evolution of envelope structures and mechanical and electrical systems since China's reform and opening-up and the beginning of its high-speed development period. Thus, this study selects 99 office buildings from this period using a full-coverage sampling method to address spatial heterogeneity. The samples cover major functional zones with high office building densities, including the Chaoyang CBD, Zhongguancun Science and Technology Park, Financial Street, and Zhongguancun Environmental Protection Industrial Park. This ensures the representation of core business districts, tech parks, and traditional financial areas. The selection criteria considered the building area, floor count, and age to reflect their gradient distribution. Data were collected from multiple sources, including field surveys, literature, government open data, and online platforms.

### 2.2. Method

Figure 1 presents the framework of the comprehensive evaluation model for low-carbon technologies developed in this study. The optimization of the technical solutions is achieved through a four-stage iterative process.

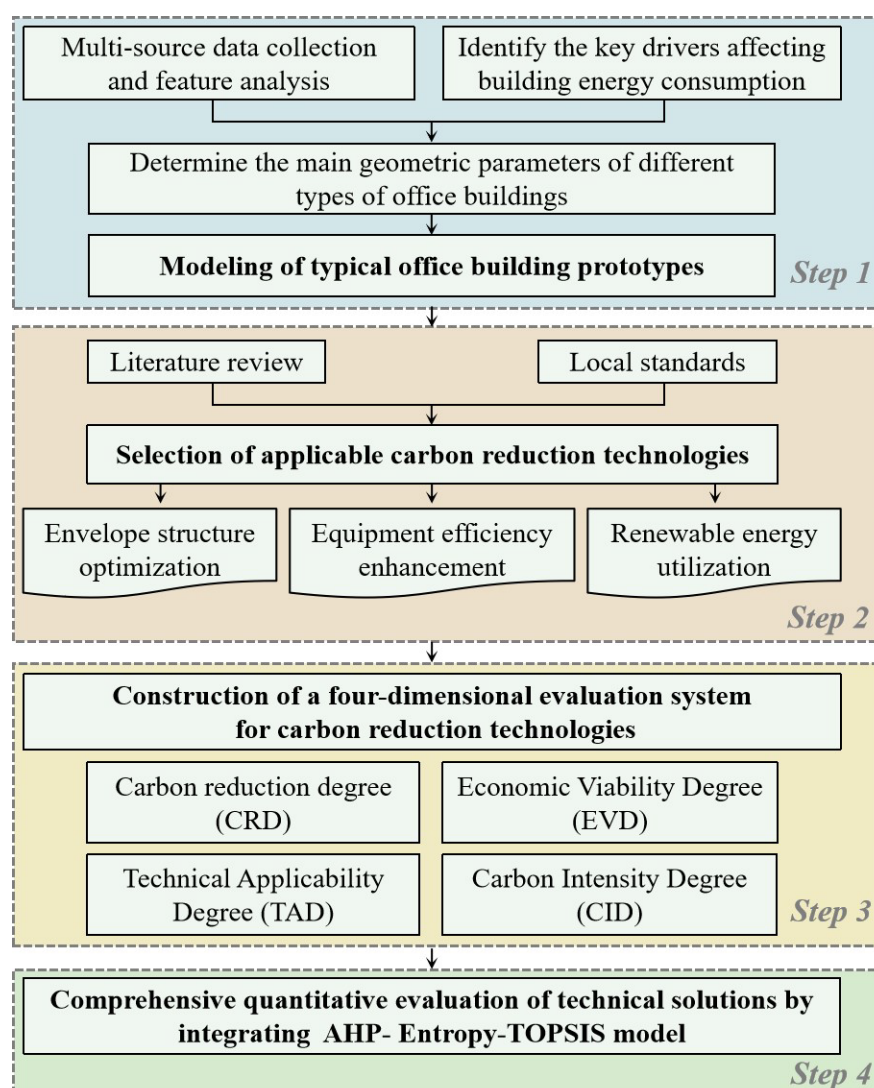
The first stage involves modeling the prototypes of typical office buildings in the region. Based on the energy consumption data from the 99 Beijing office buildings and factors related to energy consumption, the cases were classified into categories. Representative examples were selected from each category to determine the main geometric parameters of the regional architectural prototypes. Parametric modeling of these prototypes was conducted using the PKPM-GBP 2024PLUS software.

The second stage focuses on selecting applicable carbon reduction technologies. A bibliometric tool, CiteSpace, was utilized to analyze 1202 pieces of literature by searching

for the term “building low-carbon technology” in the Web of Science (WoS) Core Collection Database published in recent decades. Clusters of technology were identified through an analysis of the keyword co-occurrence network. Based on the clustering results derived from the knowledge graph, a comprehensive list of 19 low-carbon building technologies—spanning three key aspects across the entire building life cycle—was compiled.

The third stage involves the construction of a multidimensional evaluation index system. An innovative four-dimensional assessment matrix is established, comprising four key indicators: carbon reduction potential, economic feasibility, applicability, and carbon emissions. This matrix serves as the foundation of the sustainability assessment framework.

The fourth stage focuses on the comprehensive quantitative evaluation of the technical solutions. A hybrid evaluation model integrating AHP, the entropy weighting method, and TOPSIS is employed to optimize the selection of technical combinations. The comprehensive weights of the evaluation indicators are determined by integrating subjective judgements from the AHP with objective data derived from the entropy weighting method. Subsequently, multidimensional indicator decision-making is achieved using the TOPSIS approach. Finally, an optimal ranking of technical solutions is generated, enabling the identification of the most appropriate low-carbon technology options for various types of office buildings.



**Figure 1.** Framework of the proposed methodology.

### 2.2.1. Modeling of Typical Office Building Prototypes

First, multi-source data collection and feature analyses were conducted. The architectural designs of the 99 selected office buildings, as well as information related to low-carbon technologies, were obtained from the database of completed projects conducted by the China Architecture Design & Research Group Co., Ltd., or through on-site investigations. The specific sources of the data are presented in Table 1. Based on this dataset—comprising 99 office buildings constructed or renovated in Beijing between 2000 and 2022—a multidimensional database was established, covering geometric parameters (shape coefficient and window-to-wall ratio), thermal performance of envelope structures (heat transfer coefficient/U-value and solar heat gain coefficient/SHGC), and energy efficiency indicators of the equipment systems (energy consumption intensity/EUI and coefficient of performance/COP).

**Table 1.** Data sources for the sample office buildings.

| Data                                               | Source                                                               |
|----------------------------------------------------|----------------------------------------------------------------------|
| Architectural geometric parameters                 | Architectural drawings and design documentation                      |
| Thermal performance of envelope structures         | List of architectural design documentation;<br>On-site Investigation |
| Energy efficiency indicators for equipment systems | List of building service equipment;<br>On-site Investigation         |
| Low-carbon technologies employed                   | Building Energy Efficiency Report;<br>On-site Investigation          |

Second, the index parameter filtering was performed. Pearson correlation analysis ( $p < 0.01$ ) and variance inflation factor testing ( $VIF < 5$ ) were employed to systematically identify the key driving factors influencing the energy consumption of buildings. Certain indicators were excluded due to their weak relevance or negligible impact on energy performance. Specifically, architectural design codes or standardized practices often impose constraints that limit variability, thereby reducing the necessity for a correlation analysis between such parameters and energy consumption. For example, according to the Chinese national standard GB50033 “Building Daylighting Design Standard” [53], office buildings are typically designed with a north-south orientation, resulting in an insignificant correlation between the azimuth and energy consumption. Similarly, the depth-to-width ratio is often stabilized by standardized column grid designs, making its contribution to the energy prediction minimal. As a result, these non-sensitive parameters were removed following statistical validation.

Finally, parametric modeling was performed. After removing irrelevant indicators and categorizing the cases based on factors associated with energy consumption, the primary geometric parameters of the representative office building prototypes were summarized and generalized for each category. These prototypes were then modeled on the PKPM platform using the minimum energy-saving requirements of typical buildings as the benchmark.

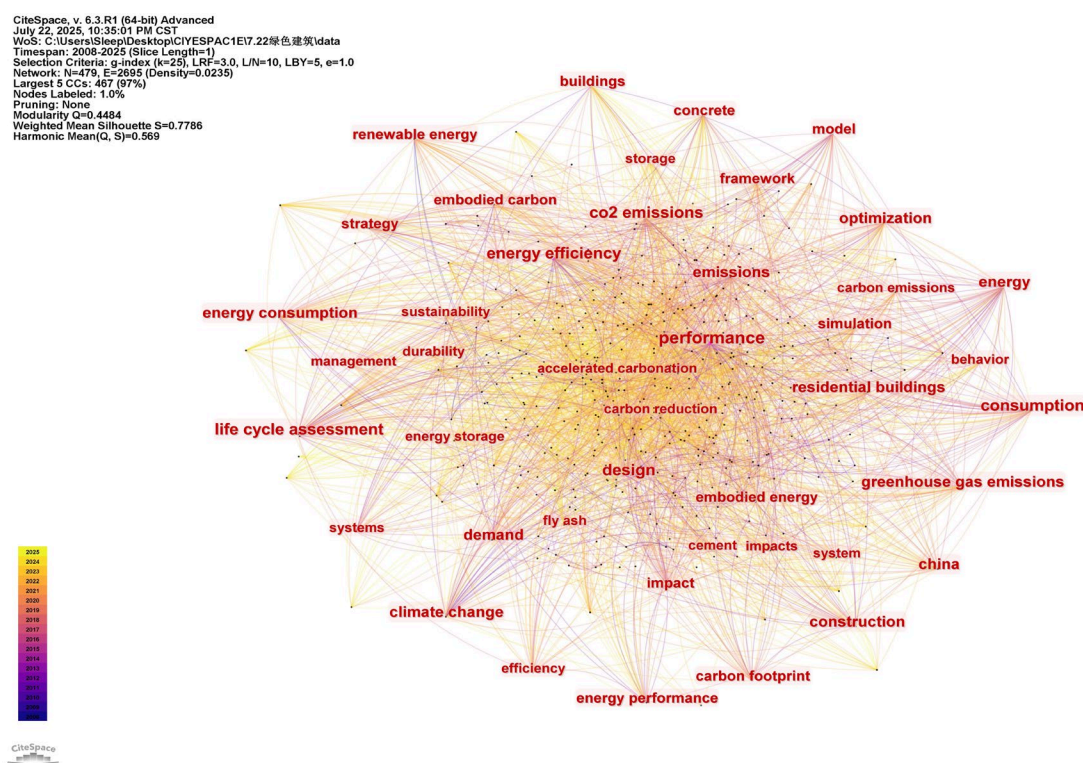
### 2.2.2. Selection of Applicable Carbon Reduction Technologies

Due to the wide variety of low-carbon technologies used in buildings and their overlapping or complementary nature, it is essential to systematically select and classify existing technologies according to different building life cycle stages. The selection process fully considered regional applicability and adhered to the local standards. A detailed explanation of the selection process is provided below.

First, this study employed the bibliometric tool CiteSpace to conduct a visual analysis of 1,202 publications from the WoS Core Collection Database covering the period from 2000 to 2025. Figure 2 presents the network of keywords pertaining to “building low-carbon technology,” and Figure 3 presents the cluster network of keywords. Eight technological clusters were identified through keyword co-occurrence network analysis. This study further classified them into three technological clusters: envelope structure optimization, equipment efficiency enhancement, and renewable-energy utilization.

Second, based on relevant standards or reports on low-carbon building technologies in Beijing, such as the local standard DB11/T 825-2021 “Beijing Green Building Evaluation Standard” [54], and “Beijing Green Building Development Report” [55], a total of 22 regionally applicable technologies were compiled and categorized according to the aforementioned three clusters.

Third, through data analysis of the application of low-carbon technologies in 99 office buildings, those among the 22 regional technologies with an adoption rate exceeding 40% were selected and confirmed as mainstream technologies applicable to office buildings in Beijing. Figure 4 presents the final list of the three clusters comprising 19 key low-carbon technologies applicable to office buildings in Beijing.



**Figure 2.** Keyword network of building low-carbon technology.

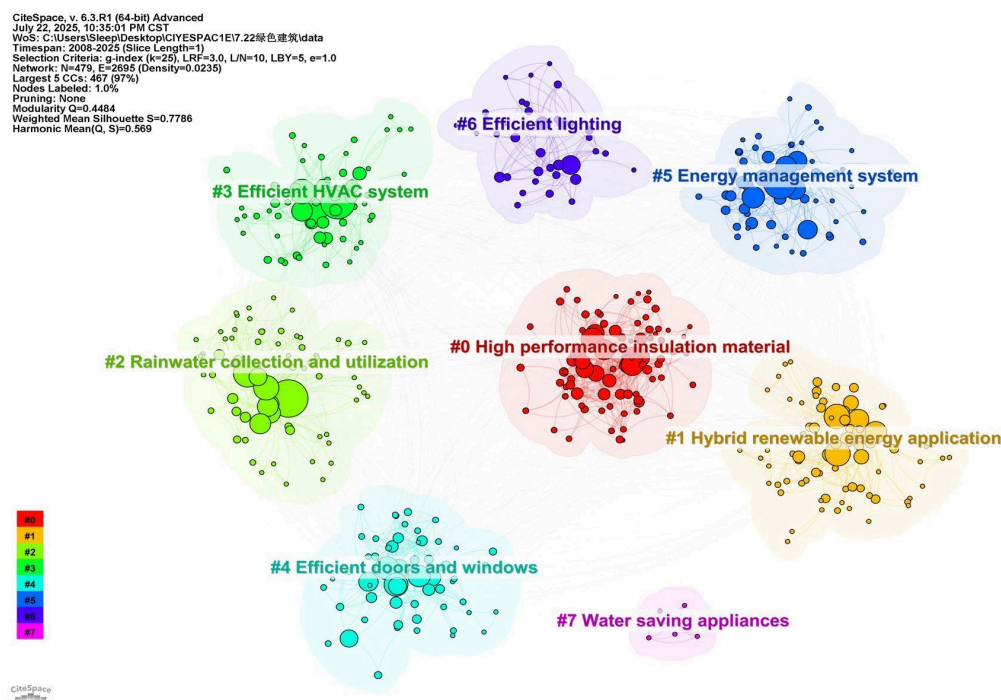


Figure 3. Keyword cluster network of building low-carbon technologies.

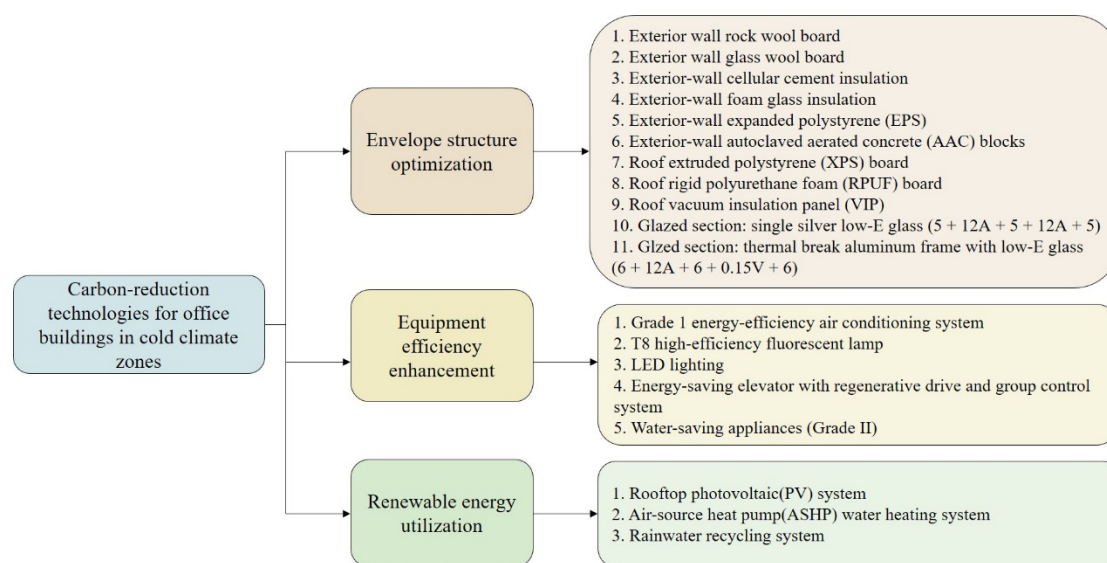


Figure 4. Nineteen key low-carbon technologies in cold climate zones.

- (1) Envelope structure optimization. The building envelope constitutes the entire physical separator between the interior and exterior environments, encompassing all external components such as walls, roofs, and windows. As a critical interface between indoor and outdoor environments, the envelope mitigates the influence of external climatic conditions on indoor thermal comfort. To achieve enhanced thermal efficiency, the following three technologies are employed:

Technology 1–6: Exterior wall insulation. In Beijing, external insulation primarily refers to exterior wall insulation technology applied to buildings. Given that Beijing is located in a cold climate zone, the exterior wall insulation must meet the national standard of 65% energy conservation for buildings. The standard specifies that the insulation performance of building exterior walls and roofs should conform to the Chinese national compulsory standard GB 50176 “Code for Thermal Design of Civil Buildings” [56]. Based

on research and statistical analysis of office buildings in Beijing. Table 2 lists the predominant usage of exterior wall materials in the region. Notably, 92% of the projects adopted external insulation composite walls, with primary materials including extruded polystyrene (XPS) board (54%), rock wool (28%), and vacuum insulation panels (18%).

**Table 2.** Statistical analysis of exterior wall materials in office buildings in Beijing.

| Material Category           | Material Name                           | Thickness (mm) | Adoption Rate |
|-----------------------------|-----------------------------------------|----------------|---------------|
| Thermal insulation material | Flame-retardant XPS board               | 40–70          | 50%           |
|                             | Cellular cement insulation board        | 50–100         | 26%           |
|                             | Foam glass insulation board             | 50–120         | 16%           |
|                             | Graphite polystyrene (GPS) board        | 70             | 5%            |
|                             | Rigid polyurethane foam (RPUF) board    | 60             | 2%            |
| Primary materials           | Reinforced concrete                     | 200            | 10%           |
|                             | Shale hollow brick                      | 200            | 12%           |
|                             | Autoclaved aerated concrete (AAC) block | 200            | 67%           |

Technology 7–9: Roof energy-saving technologies. In Beijing, most green buildings utilize an inverted roof design for roof insulation [57]. Based on research and statistical data from office buildings in Beijing, Table 3 lists the current applications of various roofing materials in such buildings.

**Table 3.** Statistical analysis of roofing materials used in office buildings in Beijing.

| Material Category            | Material Name             | Thickness (mm) | Adoption Rate |
|------------------------------|---------------------------|----------------|---------------|
| Thermal insulation materials | Flame-retardant XPS board | 70–100         | 65%           |
|                              | RPUF board                | 100            | 23%           |
|                              | GPS board                 | 100            | 12%           |
| Primary materials            | Reinforced concrete       | 120            | -             |

Technology 10–11: Optimization of exterior window performance. All green building projects under investigation adopted thermal-break aluminum double-glazed glass for windows. Specifically, approximately 65% of green residential buildings utilized double-glazed glass (model: 6 + 12a + 6), and about 30% of office buildings employed the (5 + 12a + 5) model, while roughly 5% adopted the (5 + 19a + 5) model. (Note: the model identification method for glass is defined as follows: glass thickness + air gap thickness + glass thickness, with all dimensions expressed in millimeters/mm.) However, to meet the current energy-saving standards, 5 mm-thick glass does not satisfy the required U-value criteria. Therefore, a thermal-break aluminum frame with low-E double-glazed glass (6 + 12A + 6) was selected. These are available in three variants based on the light transmittance: low, medium, and high.

- (2) Equipment efficiency enhancement. Energy-saving equipment and systems include air conditioning, lighting, hot water supply, and other related devices. In the Beijing area, energy-saving building designs must be tailored to local climatic conditions. While enhancing the heating and ventilation performance, these designs ensure both the indoor thermal environment and energy efficiency of air conditioning and lighting systems. Specifically, four technologies are presented as follows:

Technology 1: Energy conservation in air conditioners. Based on research and statistical data, all projects consistently consider the placement of outdoor units to ensure their optimal positioning. Openable aluminum alloy louvers are installed at these outdoor unit

locations, serving multiple purposes, such as shading, ventilation, protection, and prevention of airflow short circuits, while preserving the aesthetic appearance of building facades. Furthermore, during transitional seasons, split air conditioners are used in conjunction with natural ventilation strategies to reduce energy consumption. Among the 99 sample buildings, 78% used Grade I energy-efficient air-conditioning systems.

**Technology 2: Green lighting.** During the investigation of office building cases, it was found that approximately 80% of the projects employ LED lights, T8 high-efficiency fluorescent lamps, or T5 triphosphor ring-shaped energy-saving lamps as light sources, all of which are equipped with energy-saving electronic ballasts. These lighting solutions (LED lights, T8 high-efficiency fluorescent lamps, and T5 triphosphor ring-shaped energy-saving lamps) are representative of mature technologies in modern energy-efficient illumination. Their energy consumption is considerably lower than that of traditional lighting devices, such as incandescent lamps, while maintaining comparable or superior lighting quality. Furthermore, 86% of the surveyed projects implemented illuminance-sensing systems combined with zonal control strategies. LED lights still account for the largest proportion of green lighting technologies, reaching 52%, while T8 high-efficiency fluorescent lamps constitute 32%.

**Technology 3: Energy conservation in electrical designs.** Research on office building cases has revealed that approximately 60% of engineering projects incorporate energy-saving elevator systems. In areas with multiple elevators, energy-efficient control measures, such as group control systems and automatic lighting shutoff technology during periods of non-use, are implemented. Furthermore, twenty-three projects have adopted SCB-type dry-type transformers, which are characterized by low losses, low noise, and high energy efficiency. These transformers meet the energy efficiency limit values specified in the Chinese national compulsory standard GB 20052 “Energy Consumption Limit Values and Energy Efficiency Evaluation Values for Three-phase Distribution Transformers” [58], and conform to the target energy efficiency requirements. The recommended long-term load rate for these transformers should not exceed 85%. Additionally, in 55% of the analyzed project cases, variable-frequency speed control pumps are utilized in the heating, ventilation, and air conditioning (HVAC) systems and water supply/drainage systems, allowing for adaptive speed regulation and energy savings in response to varying load demands.

**Technology 4: Water-saving appliances.** The water supply equipment and its accessories fully comply with the requirements of two Chinese national standards, namely GB/T 31436-2015 “Water-Saving Domestic Water Appliances” [59] and GB/T 8870-2011 “General Technical Conditions for Water-saving Products” [60]. In accordance with the green building requirements, appliances with water efficiency grades I or II were selected. Toilets should be equipped with a trap seal of at least 50 mm. The adoption rate of water-saving appliances exceeds 75%.

(3) **Renewable energy utilization.** The utilization of renewable energy serves as a critical technical approach to achieving carbon neutrality in the construction sector, particularly through the application of solar and geothermal energy in buildings. These technologies primarily include solar thermal energy, photovoltaic (PV) systems or building-integrated photovoltaics (BIPV), ground-source heat pumps, air-source heat pumps, and wind energy [3]. Although Beijing’s wind resources are technically exploitable, a distinct administrative barrier specific to wind energy has effectively suppressed its adoption in urban office projects; therefore, wind energy was excluded from the renewable energy mix considered in this analysis. Currently, three carbon reduction technologies related to renewable energy utilization are mainly applicable in Beijing.

Technology 1: Rooftop PV systems. Solar PV power generation typically utilizes the roof space of a building. By installing PV panels, sunlight is converted into electrical energy and then integrated into the power grid for practical use. Market research on general flat roofs indicates that PV power generation has not yet been widely adopted in office buildings. However, with strong policy support, there is still sufficient potential for further development in this area. PV technology is particularly suitable as a renewable energy source due to its stable market growth and declining costs [61]. Beyond power generation, BIPV modules often provide at least one additional function, such as insulation, wind and rain protection, and shading [62]. Studies have shown that net-zero energy consumption can be achieved by combining PV panels with solar heating systems [63]. Among the surveyed buildings, 63% had rooftop PV installations.

Technology 2: Air-source heat pump (ASHP) system. An air-source heat pump system is a renewable energy technology that uses ambient air to generate usable heat. By consuming a relatively small amount of electricity, the system extracts low-grade thermal energy from the surrounding air and converts it into high-grade thermal energy [6]. Air-source heat pumps are widely applied in building HVAC systems, particularly in cold regions and areas characterized by hot summers and cold winters, where their performance is especially effective. At present, ASHPs are extensively used in residential and commercial air conditioning, water heating, and space heating applications. They offer significant advantages in terms of environmental sustainability, energy efficiency, and reduced operational costs. Due to their lower installation costs and ease of integration into existing buildings compared with other heat pump technologies, air-water heat pumps are increasingly regarded as practical and viable alternative solutions [29]. Among the surveyed buildings, the ASHP adoption rate accounted for 43%.

Technology 3: Rainwater recycling and utilization system. The rainwater recycling and utilization system involves collecting and storing available rainwater resources, which are then subjected to simple treatment before reuse in residential areas or other regions. This not only conserves water resources but also reduces environmental pollution. Recycled rainwater can be used for flushing toilets, irrigating green spaces, and other non-potable purposes. It offers the dual advantages of water conservation and environmental protection, while promoting sustainable development. With the implementation of sponge city initiatives (a national strategy in China aimed at enhancing urban resilience by mimicking natural hydrology through an integrated “green-grey” infrastructure for stormwater infiltration, retention, and reuse) [64], rainwater recycling and utilization technologies have been increasingly promoted and applied, particularly in office buildings, where collected rainwater can serve as a water source for flushing toilets. Among the surveyed buildings, the adoption rate of stormwater recycling and utilization systems was 53%.

### 2.2.3. Construction of a Four-Dimensional Evaluation System for Carbon Reduction Technologies

This study comprehensively considers aspects such as high carbon reduction benefits, optimal cost performance, and convenience for actual construction and subsequent maintenance, and builds a four-dimensional evaluation system to quantitatively evaluate the aforementioned carbon reduction technologies. The assessment is conducted based on four indicators: Carbon Reduction Degree (CRD), Economic Viability Degree (EVD), Technical Applicability Degree (TAD), and Carbon Intensity Degree (CID). The quantification methods for each dimension are as follows:

- (1) Carbon Reduction Degree (CRD)

The CRD index quantifies the overall carbon reduction performance of a building after implementing a specific technology. Derived from the parameters of typical buildings, the CRD index is calculated as the percentage of the annual carbon reduction achieved by applying each technology to the annual carbon emissions of a baseline scenario without any carbon reduction measures, as defined in Equation (1).

$$CRD_n = \frac{\Delta CE_n}{CE_{base}} \times 100\%, (n = 1, 2, 3, \dots, 19) \quad (1)$$

Among them,  $CE_{base}$  refers to the annual carbon emissions (in kg) of the building baseline scenario without any carbon reduction technologies;  $\Delta CE_n$  represents the annual carbon reductions (in kg) achieved by implementing carbon reduction technology  $n$ , which are obtained through simulation. The PKPM software can conduct comprehensive simulations of energy consumption and economic analyses for building heating, cooling, lighting, and daylighting systems, among others; thus, it has been widely adopted in building carbon emission research.

#### (2) Economic Viability Degree (EVD)

Following consultations with manufacturers and verification by experts, cost data for the application of multiple technologies under general working conditions were collected. Subsequently, three types of typical office buildings were selected for the case studies. Their specific architectural parameters, appliance, and equipment configurations were described in detail, allowing the calculation of the incremental cost per unit building area for each carbon reduction technology. The unit prices of various building materials were directly obtained through market comparison research. For energy-saving doors and windows, the incremental cost per unit area is determined by dividing the increased price per unit area by the window-to-floor ratio of the corresponding building type. Based on these incremental cost data, the EVD index is calculated using Equation (2).

$$EVD_n = \frac{EC_n}{MAX_{EC}} \times 100\%, (n = 1, 2, 3, \dots, 19) \quad (2)$$

Among them,  $EVD_n$  represents the EVD of the 19 carbon reduction technologies;  $EC_n$  refers to the incremental cost per unit area (yuan/m<sup>2</sup>) associated with the corresponding technology, and  $MAX_{EC}$  denotes the maximum incremental cost per unit area across all studied technologies. A higher value of this economic indicator indicates that a greater upfront investment is required for the implementation of the technology.

#### (3) Technical Applicability Degree (TAD)

A semi-structured interview method was used for the questionnaire survey. A specific engineering case was selected, and based on the key technologies and materials involved in this case, investigations were conducted from multiple aspects, including technological maturity, regional climate, and policy applicability. When scoring by experts, in addition to the above factors, various uncertainties that affect the adoption and performance of low-carbon technologies, such as future subsidy revisions, component efficiency gains, module price trajectories, and the pace of green-power grid integration, have also been taken into account in the experts' scoring. Subsequently, the TAD of the building carbon reduction technology was calculated using Equation (3).

$$TAD_n = \frac{\sum_{i=1}^k \omega_i \times S_{in}}{\sum_{i=1}^k \omega_i} \times 100\%, (n = 1, 2, 3, \dots, 19) \quad (3)$$

Among them,  $TAD_n$  represents the applicability degree of the 19 carbon reduction technologies;  $S_{in}$  denotes the expert score (on a scale ranging from 1 to 9), and  $\omega_i$  indicates the index weight determined by the AHP method.

#### (4) Carbon Intensity Degree (CID)

The annual carbon emissions of three typical office building prototypes were obtained through simulations and calculations using PKPM software. Carbon emissions intensity (CEI) was calculated using Equation (4). After normalization, the data were standardized to a uniform scale (0–1), resulting in the CID index, as shown in Equation (5).

$$CEI_n = \frac{CE_n}{A_i} \quad (4)$$

$$CID_n = \frac{CEI_n - CEI_{min}}{CEI_{max} - CEI_{min}} \times 100\%, (n = 1, 2, 3, \dots, 19) \quad (5)$$

Among them,  $CID_n$  represents the carbon emission intensity index of three typical office building prototypes after applying various carbon reduction technologies, and  $CEI_n$  refers to the annual carbon emission intensity (in kg/m<sup>2</sup>) of the building after adopting carbon reduction technology  $n$ .  $CEI_{max}$  and  $CEI_{min}$  denote the maximum and minimum values, respectively, of the annual carbon emission intensities of the building after adopting carbon reduction technology  $n$ .

#### 2.2.4. Comprehensive Quantitative Decision-Making for Technical Solutions

The MCDM-based system evaluation framework supports comprehensive consideration of multiple dimensions and factors. By ranking various technical solutions, the optimal solution for the low-carbon design and renovation of buildings can be identified [65,66]. Common MCDM methods include AHP, TOPSIS, WASPAS, COPRAS, ELECTRE, PROMETHEE, VIKOR, EDAS, WSM, and so on [27,67,68]. Among these, AHP can be used to evaluate the weights of preference criteria, enabling the consideration of preferences from multiple stakeholders [69]. The Entropy Weight Method can effectively eliminate the possible cognitive biases in the AHP, and its principle is to objectively extract distinguishable information from quantitative data. The TOPSIS method determines the solution to a multi-objective decision-making problem based on the decision-maker's priorities and specific weights assigned to each criterion. It can achieve a simple and clear ranking while avoiding the common ranking reversal problem in additional models. The AHP and TOPSIS methods are often combined to evaluate the comprehensive performance of different technical solutions [70–72]. This study integrates AHP-Entropy-TOPSIS and considers diverse stakeholder preferences to determine the preferred scheme for low-carbon technology.

According to the constructed four-dimensional carbon reduction technology evaluation system, the improved AHP-Entropy-TOPSIS comprehensive evaluation model is applied to quantitatively optimize the technology combination schemes. Based on the results of the quantitative assessment, differentiated carbon reduction technical paths are proposed for different types of office buildings. The specific steps are as follows.

Firstly, the combined weighting method is employed to determine the comprehensive weight. This approach integrates both subjective and objective weighting by combining the AHP and entropy weight methods to determine the weights of the comprehensive indicators. In AHP, a judgment matrix is constructed using expert questionnaires, and weights are assigned based on subjective evaluations [73]. Eight experts, including four construction practitioners and four university professors, participated in this study by completing a fuzzy Delphi expert questionnaire. All questionnaires were collected and validated. The entropy weight method, on the other hand, determines the weights based on the inherent distribution of the data. The integration of subjective and objective weights is shown in Equation (6). The final weights of each of the four-dimensional indicators were obtained by taking a weighted average of the AHP results with those from the entropy weight method and balancing expert experience with data distribution characteristics.

$$\omega_j = \alpha \times \omega_{AHP} + (1 - \alpha) \times \omega_{Entropy} \quad (6)$$

Among them,  $\alpha = 0.5$  was determined by a sensitivity analysis.

Secondly, the comprehensive decision-making of multidimensional indicators is achieved through the application of TOPSIS, and the superiority and inferiority grades of the technology are quantified using Euclidean distance. The scheme closest to the ideal solution was selected by calculating the distances of each scheme from the ideal and negative ideal solutions. A standardized decision matrix is constructed (Equation (7)), and the distances between the positive and negative ideal solutions are calculated (Equation (8)). The comprehensive score of each scheme is determined using Equation (9). Based on the optimal ranking derived from these comprehensive scores, three types of differentiated carbon-reduction technology paths are established for typical office buildings.

$$\gamma_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}} \quad (7)$$

$$D_i^+ = \sqrt{\sum_{j=1}^n \omega_j \times (\gamma_{ij} - \gamma_j^+)^2} \quad (8)$$

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-} \quad (9)$$

### 3. Results and Analyses

#### 3.1. Modeling Results of Typical Regional Architectural Prototypes

Based on the research data of 99 office buildings in the Beijing area and factors related to energy consumption, the cases were classified as follows: representative examples were selected from each category, and nine key energy consumption influencing factors were identified through the Pearson correlation coefficient matrix ( $|r| > 0.6$ ), namely the primary geometric parameters of typical building prototypes (Table 4). Using the minimum energy-saving requirements of these typical buildings as a benchmark, the corresponding construction methods for the envelope structure and the parameters of the basic equipment were determined (Table 5), and prototype modeling of the three types of office buildings was completed using PKPM. These include low-rise, mid-rise, and high-rise structures (Figure 5).

**Table 4.** Basic information on the simulated buildings.

| Basic Information                     |       | Low-Rise<br>(Prototype 1) | Mid-Rise<br>(Prototype 2) | High-Rise<br>(Prototype 3) |
|---------------------------------------|-------|---------------------------|---------------------------|----------------------------|
| Building area (m <sup>2</sup> )       |       | 2220.8                    | 17376.0                   | 47,128.4                   |
| Height (m)                            |       | 24                        | 50                        | 96.6                       |
| Surface area (m <sup>2</sup> )        |       | 2671.4                    | 9169                      | 19,572.3                   |
| Standard floor area (m <sup>2</sup> ) |       | 444.15                    | 1449                      | 1448                       |
| Standard floor height (m)             |       | 4.8                       | 4.16                      | 4.2                        |
| Number of building floors             |       | 5                         | 12                        | 23                         |
| Shape coefficient                     |       | 0.25                      | 0.13                      | 0.1                        |
| Appearance coefficient                |       | 1.2                       | 0.53                      | 0.42                       |
| Window-to-wall ratio                  | East  | 0.2                       | 0.2                       | 0.21                       |
|                                       | South | 0.19                      | 0.24                      | 0.27                       |
|                                       | West  | 0.2                       | 0.2                       | 0.21                       |
|                                       | North | 0.19                      | 0.24                      | 0.27                       |

Table 5. Model parameter information.

| Content                                        | Specific Approach                                                                                    | Specification      |
|------------------------------------------------|------------------------------------------------------------------------------------------------------|--------------------|
| Roof structure                                 | Asbestos cement roof                                                                                 | 20 mm              |
|                                                | Fine aggregate concrete                                                                              | 40 mm              |
|                                                | Waterproof membrane                                                                                  | 2 mm               |
|                                                | Cement mortar                                                                                        | 20 mm              |
|                                                | XPS board                                                                                            | 75 mm              |
|                                                | Lightweight aggregate concrete                                                                       | 30 mm              |
| Exterior wall                                  | Reinforced concrete                                                                                  | 120 mm             |
|                                                | Cement mortar                                                                                        | 10 mm              |
|                                                | Rock wool board                                                                                      | 50 mm              |
|                                                | AAC block, density grade B06                                                                         | 200 mm             |
|                                                | Cement mortar                                                                                        | 10 mm              |
| Exterior window                                | Thermal break aluminum frame with multi-chamber structure, glazed section: low-E glass (6 + 12A + 6) |                    |
| COP of a Multi-split system                    | -                                                                                                    | 4.1                |
| Lighting power density (LPD)—incandescent lamp | -                                                                                                    | 8 w/m <sup>2</sup> |

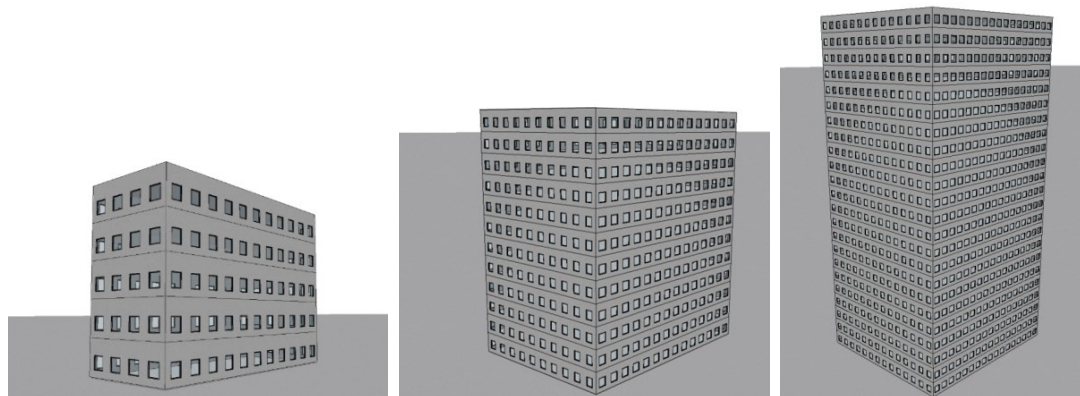


Figure 5. Three typical architectural prototype models.

### 3.2. Four-Dimensional Evaluation Results of Key Carbon Reduction Technologies

This study takes the aforementioned three types of typical office buildings in Beijing as research objects and conducts a comprehensive evaluation of 19 carbon reduction technologies based on a four-dimensional evaluation system. The performance differences and internal contradictions among these technologies are revealed by the scores of the four-dimensional indicators. The following section presents an analysis of each of these dimensions.

#### (1) CRD evaluation results

Rooftop PV systems (CRD = 0.686) and LED lighting (CRD = 0.800) demonstrated the best performance. Among these, LED lighting achieved a 100% carbon reduction effect in mid-rise and high-rise structures (prototypes 2 and 3, see Table 6). Energy-saving elevators (CRD = 0.043) and water-saving appliances (CRD = 0.004) contributed the least to carbon reduction due to their limited application scope. Notably, the CRD of the first-level energy-efficient air conditioning system in mid-rise structures (prototype 2) reached 0.257, which is significantly higher than that of envelope structure optimization technology (average CRD =  $0.026 \pm 0.003$ ), indicating that improving equipment energy efficiency is a key pathway for achieving deep carbon reduction.

**Table 6.** Normalized CRD results.

| Serial | Technology Name                                                                         | Low-Rise<br>Prototype 1 | Mid-Rise<br>Prototype2 | High-Rise<br>Prototype 3 | Average<br>Value |
|--------|-----------------------------------------------------------------------------------------|-------------------------|------------------------|--------------------------|------------------|
| 1      | Exterior wall rock wool board                                                           | 0.029                   | 0.029                  | 0.023                    | 0.027            |
| 2      | Exterior wall glass wool board                                                          | 0.029                   | 0.027                  | 0.021                    | 0.026            |
| 3      | Exterior-wall cellular cement insulation board                                          | 0.029                   | 0.028                  | 0.022                    | 0.026            |
| 4      | Exterior-wall foam glass insulation board                                               | 0.030                   | 0.029                  | 0.022                    | 0.027            |
| 5      | Exterior-wall expanded polystyrene (EPS) board                                          | 0.025                   | 0.022                  | 0.017                    | 0.021            |
| 6      | Exterior-wall AAC blocks                                                                | 0.021                   | 0.021                  | 0.013                    | 0.018            |
| 7      | Roof XPS board                                                                          | 0.004                   | 0.006                  | 0.005                    | 0.005            |
| 8      | Roof RPUF board                                                                         | 0.003                   | 0.005                  | 0.004                    | 0.004            |
| 9      | Roof vacuum insulation panel (VIP)                                                      | 0.003                   | 0.004                  | 0.004                    | 0.004            |
| 10     | Glazed section: single silver low-E glass (5 + 12A + 5 + 12A + 5)                       | 0.026                   | 0.037                  | 0.032                    | 0.032            |
| 11     | Glazed section: thermal break aluminum frame with low-E glass (6 + 12A + 6 + 0.15V + 6) | 0.054                   | 0.069                  | 0.059                    | 0.061            |
| 12     | Grade 1 energy-efficiency air conditioning system                                       | 0.154                   | 0.257                  | 0.236                    | 0.216            |
| 13     | LED lighting                                                                            | 0.401                   | 1.000                  | 1.000                    | 0.800            |
| 14     | T8 high-efficiency fluorescent lamp                                                     | 0.268                   | 0.667                  | 0.667                    | 0.534            |
| 15     | Energy-saving elevator with regenerative drive and group control system                 | 0.024                   | 0.040                  | 0.066                    | 0.043            |
| 16     | Water-saving appliances (Grade II)                                                      | 0.002                   | 0.004                  | 0.004                    | 0.004            |
| 17     | Rooftop PV system                                                                       | 1.000                   | 0.672                  | 0.386                    | 0.686            |
| 18     | ASHP water heating system                                                               | 0.075                   | 0.099                  | 0.093                    | 0.089            |
| 19     | Rainwater recycling system                                                              | 0.001                   | 0.001                  | 0.000                    | 0.001            |

**(2) EVD evaluation results**

The rainwater recycling system leads with an ECD of 0.665; however, its economic advantage is primarily driven by policy subsidies (see Table 7). Passive technologies, such as vacuum insulation panels (ECD = 0.004) and extruded polystyrene boards (ECD = 0.001), exhibit the lowest incremental costs. In contrast, autoclaved aerated concrete blocks (ECD = 0.152) face notable economic disadvantages due to construction complexity. A comparative analysis reveals a common cost paradox in equipment technologies: although the first-level energy efficiency air conditioner (ECD = 0.859) demonstrates significant carbon reduction performance, its incremental cost is as much as 16.5 times higher than that of LED lighting.

**Table 7.** Normalized EVD results.

| Serial | Technology Name                                                                        | Low-Rise<br>Prototype 1 | Mid-Rise<br>Prototype2 | High-Rise<br>Prototype 3 | Average<br>Value |
|--------|----------------------------------------------------------------------------------------|-------------------------|------------------------|--------------------------|------------------|
| 1      | Exterior wall rock wool board                                                          | 0.021                   | 0.016                  | 0.013                    | 0.017            |
| 2      | Exterior wall glass wool board                                                         | 0.030                   | 0.024                  | 0.019                    | 0.025            |
| 3      | Exterior-wall cellular cement insulation board                                         | 0.045                   | 0.036                  | 0.029                    | 0.037            |
| 4      | Exterior-wall foam glass insulation board                                              | 0.048                   | 0.038                  | 0.030                    | 0.039            |
| 5      | Exterior-wall EPS board                                                                | 0.030                   | 0.023                  | 0.019                    | 0.024            |
| 6      | Exterior-wall AAC blocks                                                               | 0.188                   | 0.148                  | 0.119                    | 0.152            |
| 7      | Roof XPS board                                                                         | 0.001                   | 0.001                  | 0.000                    | 0.001            |
| 8      | Roof RPUF board                                                                        | 0.009                   | 0.006                  | 0.003                    | 0.006            |
| 9      | Roof VIP                                                                               | 0.006                   | 0.004                  | 0.002                    | 0.004            |
| 10     | Glazed section: single silver low-E glass (5+ 12A + 5+ 12A + 5)                        | 0.103                   | 0.122                  | 0.111                    | 0.112            |
| 11     | Glazed section: thermal break aluminum frame with low-E glass (6 + 12A + 6 + 0.15V+ 6) | 0.267                   | 0.316                  | 0.287                    | 0.290            |
| 12     | Grade 1 energy-efficiency air conditioning system                                      | 0.578                   | 1.000                  | 1.000                    | 0.859            |
| 13     | LED lighting                                                                           | 0.035                   | 0.060                  | 0.060                    | 0.052            |
| 14     | T8 high-efficiency fluorescent lamp                                                    | 0.012                   | 0.021                  | 0.021                    | 0.018            |
| 15     | Energy-saving elevator with regenerative drive and group control system                | 0.031                   | 0.053                  | 0.053                    | 0.046            |
| 16     | Water-saving appliances (Grade II)                                                     | 0.008                   | 0.014                  | 0.014                    | 0.012            |
| 17     | Rooftop PV system                                                                      | 0.405                   | 0.250                  | 0.152                    | 0.269            |
| 18     | ASHP water heating system                                                              | 0.035                   | 0.061                  | 0.061                    | 0.053            |
| 19     | Rainwater recycling system                                                             | 1.000                   | 0.618                  | 0.376                    | 0.665            |

### (3) TAD evaluation results

Mature technologies exhibit a polarization trend (see Table 8). The highly applicable technology group ( $TAD \geq 0.9$ ) includes rock wool insulation boards, standard low-E windows, and LED lighting. These technologies have established a large-scale application ecosystem. The low-applicable technology group ( $TAD \leq 0.3$ ) consists of ASHP and BIPV systems, which are primarily constrained by installation conditions and the extent of policy alignment. Although the rainwater recycling system ( $TAD = 0.3$ ) aligns with the development direction of sponge cities, its adoption is limited by annual rainfall fluctuations in Beijing ( $585 \pm 132$  mm/a), resulting in an actual adoption rate of less than 15%.

**Table 8.** Normalized TAD results.

| Serial | Technology Name                                                                         | Low-Rise<br>Prototype 1 | Mid-Rise<br>Prototype2 | High-Rise<br>Prototype 3 | Average<br>Value |
|--------|-----------------------------------------------------------------------------------------|-------------------------|------------------------|--------------------------|------------------|
| 1      | Exterior wall rock wool board                                                           | 0.900                   | 0.900                  | 0.900                    | 0.900            |
| 2      | Exterior wall glass wool board                                                          | 0.300                   | 0.300                  | 0.300                    | 0.300            |
| 3      | Exterior-wall cellular cement insulation board                                          | 0.300                   | 0.300                  | 0.300                    | 0.300            |
| 4      | Exterior-wall foam glass insulation board                                               | 0.300                   | 0.300                  | 0.300                    | 0.300            |
| 5      | Exterior-wall EPS board                                                                 | 0.600                   | 0.600                  | 0.600                    | 0.600            |
| 6      | Exterior-wall AAC blocks                                                                | 0.900                   | 0.900                  | 0.900                    | 0.900            |
| 7      | Roof XPS board                                                                          | 0.900                   | 0.900                  | 0.900                    | 0.900            |
| 8      | Roof RPUF board                                                                         | 0.600                   | 0.600                  | 0.600                    | 0.600            |
| 9      | Roof VIP                                                                                | 0.300                   | 0.300                  | 0.300                    | 0.300            |
| 10     | Glazed section: single silver low-E glass (5 + 12A + 5 + 12A + 5)                       | 0.900                   | 0.900                  | 0.900                    | 0.900            |
| 11     | Glazed section: thermal break aluminum frame with low-E glass (6 + 12A + 6 + 0.15V + 6) | 0.300                   | 0.300                  | 0.300                    | 0.300            |
| 12     | Grade 1 energy-efficiency air conditioning system                                       | 0.600                   | 0.600                  | 0.600                    | 0.600            |
| 13     | LED lighting                                                                            | 0.900                   | 0.900                  | 0.900                    | 0.900            |
| 14     | T8 high-efficiency fluorescent lamp                                                     | 0.900                   | 0.900                  | 0.900                    | 0.900            |
| 15     | Energy-saving elevator with regenerative drive and group control system                 | 0.600                   | 0.600                  | 0.600                    | 0.600            |
| 16     | Water-saving appliances (Grade II)                                                      | 0.900                   | 0.900                  | 0.900                    | 0.900            |
| 17     | Rooftop PV system                                                                       | 0.600                   | 0.600                  | 0.600                    | 0.600            |
| 18     | ASHP water heating system                                                               | 0.300                   | 0.300                  | 0.300                    | 0.300            |
| 19     | Rainwater recycling system                                                              | 0.300                   | 0.300                  | 0.300                    | 0.300            |

**(4) CID evaluation results**

Among the peripheral protection technologies, thermal-break aluminum composite windows (CID = 0.010) have been identified as a carbon emission hotspot, primarily due to their excessively high embodied carbon (14.3 kgCO<sub>2</sub>/m<sup>2</sup>) during the production stage (see Table 9). In contrast, XPS board roofs (CID = 0.990) demonstrate superior low-carbon performance across their entire life cycle, which is attributed to their relatively low carbon emission ratio during the construction phase (< 5%).

**Table 9.** Normalized CID results.

| Serial | Technology Name                | Low-Rise<br>Prototype 1 | Mid-Rise<br>Prototype2 | High-Rise<br>Prototype 3 | Average Value |
|--------|--------------------------------|-------------------------|------------------------|--------------------------|---------------|
| 1      | Exterior wall rock wool board  | 0.878                   | 0.919                  | 0.928                    | 0.908         |
| 2      | Exterior wall glass wool board | 0.935                   | 0.957                  | 0.962                    | 0.951         |

|    |                                                                                         |       |       |       |       |
|----|-----------------------------------------------------------------------------------------|-------|-------|-------|-------|
| 3  | Exterior-wall cellular cement insulation board                                          | 0.699 | 0.801 | 0.822 | 0.774 |
| 4  | Exterior-wall foam glass insulation board                                               | 0.805 | 0.871 | 0.885 | 0.854 |
| 5  | Exterior-wall EPS board                                                                 | 0.938 | 0.959 | 0.963 | 0.954 |
| 6  | Exterior-wall AAC blocks                                                                | 0.179 | 0.456 | 0.515 | 0.383 |
| 7  | Roof XPS board                                                                          | 0.991 | 0.994 | 0.994 | 0.993 |
| 8  | Roof RPUF board                                                                         | 0.980 | 0.950 | 0.956 | 0.962 |
| 9  | Roof VIP                                                                                | 0.993 | 0.981 | 0.983 | 0.986 |
| 10 | Glazed section: single silver low-E glass (5 + 12A + 5 + 12A + 5)                       | 0.167 | 0.167 | 0.167 | 0.167 |
| 11 | Glazed section: thermal break aluminum frame with low-E glass (6 + 12A + 6 + 0.15V + 6) | 0.001 | 0.001 | 0.001 | 0.001 |
| 12 | Grade 1 energy-efficiency air conditioning system                                       | 0.878 | 0.919 | 0.928 | 0.908 |
| 13 | LED lighting                                                                            | 0.878 | 0.919 | 0.928 | 0.908 |
| 14 | T8 high-efficiency fluorescent lamp                                                     | 0.878 | 0.919 | 0.928 | 0.908 |
| 15 | Energy-saving elevator with regenerative drive and group control system                 | 0.878 | 0.919 | 0.928 | 0.908 |
| 16 | Water-saving appliances (Grade II)                                                      | 0.878 | 0.919 | 0.928 | 0.908 |
| 17 | Rooftop PV system                                                                       | 0.878 | 0.919 | 0.928 | 0.908 |
| 18 | ASHP water heating system                                                               | 0.878 | 0.919 | 0.928 | 0.908 |
| 19 | Rainwater recycling system                                                              | 0.878 | 0.919 | 0.928 | 0.908 |

### 3.3. Low Carbon Technologies for Different Types of Office Buildings

Based on the four-dimensional index evaluation results of 19 technologies across three typical office building prototypes, the improved TOPSIS method was applied to calculate the comprehensive scores and rankings of the low-carbon technologies for each building type. Accordingly, differentiated low-carbon technology strategies are proposed for different categories of office buildings.

#### 3.3.1. Analysis of Carbon Reduction Technologies for Low-Rise Office Buildings

Based on the four-dimensional assessment results of the comprehensive ranking and classification of low-carbon technologies (see Tables 10 and 11), for low-rise office buildings (Prototype 1), applicable carbon reduction technologies can be classified into three categories: improving the thermal performance of the envelope structure, adopting high-efficiency equipment, and recycling and utilizing energy resources. Among these categories, energy resource recycling and utilization ranks first. Rooftop PV power generation has emerged as the most effective technology due to its superior carbon reduction (energy conservation) performance. Second is the technology category of high-efficiency equipment, where LED lighting stands out as the optimal choice because of its exceptional carbon reduction (energy conservation) performance and broad applicability. In the category of enhancing the thermal performance of envelope structures, the application of thermally broken aluminum frames with low-E double-glazed glass (6 + 12A + 6 + 0.15V + 6) significantly improves the thermal performance of transparent envelope components, making it the preferred technology in this category due to its excellent energy-saving performance and relatively low unit-area carbon emissions. Additionally, the use of double-glazed laminated glass (single silver low-E 5 + 12A + 5 + 12A + 5) and exterior-wall AAC blocks also

demonstrates outstanding performance in terms of applicability and carbon footprint control.

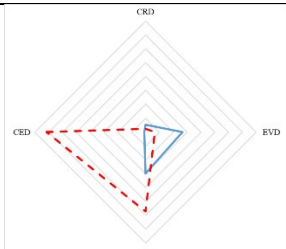
Therefore, among the three categories of technologies—rooftop PV systems, LED lighting, and low-E double-glazed laminated glass (6+12A+6+0.15V+6) with thermally broken aluminum frames— these should be prioritized for energy-saving renovations in similar office buildings.

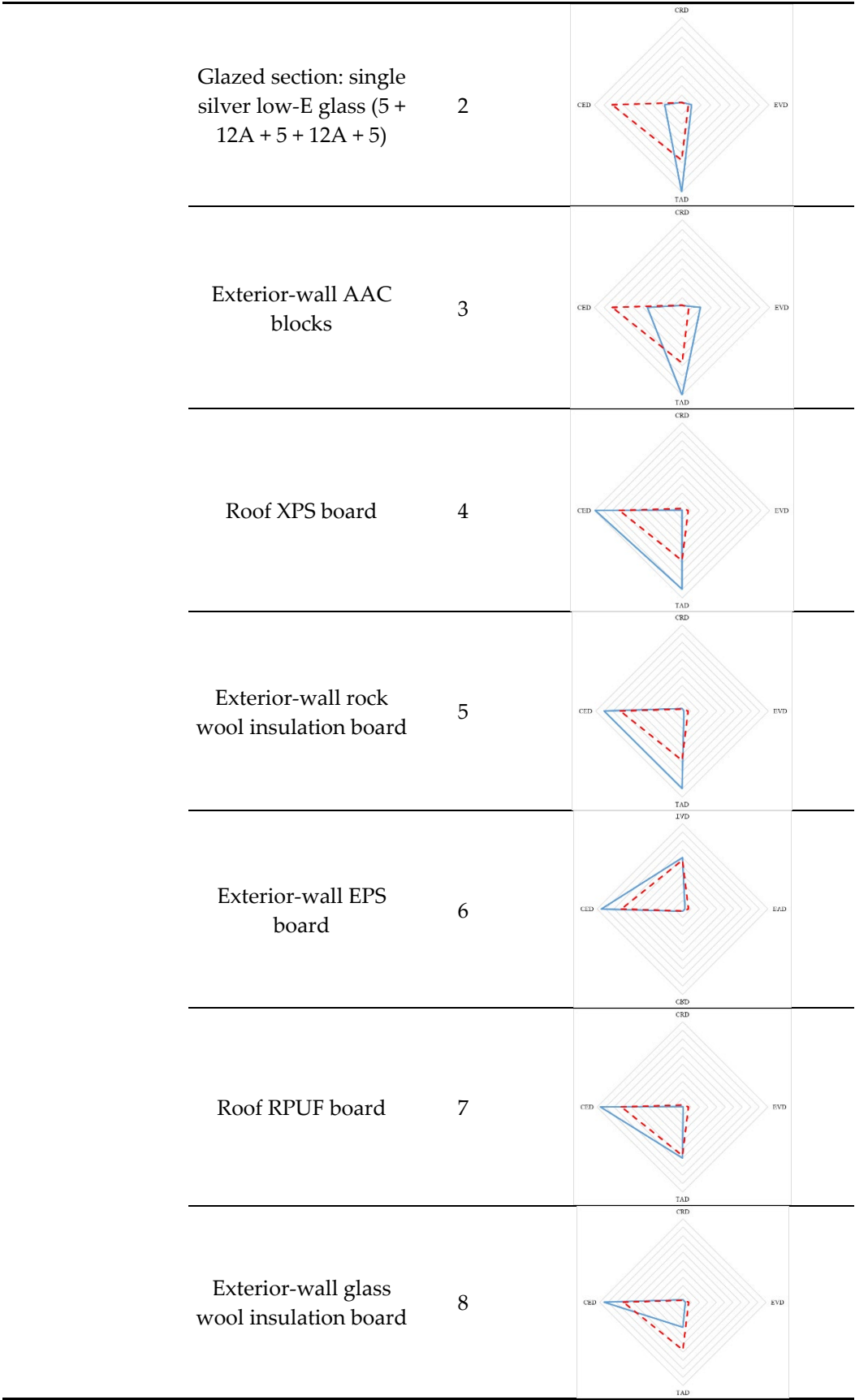
**Table 10.** Comprehensive score and ranking of low-carbon technologies for Prototype 1.

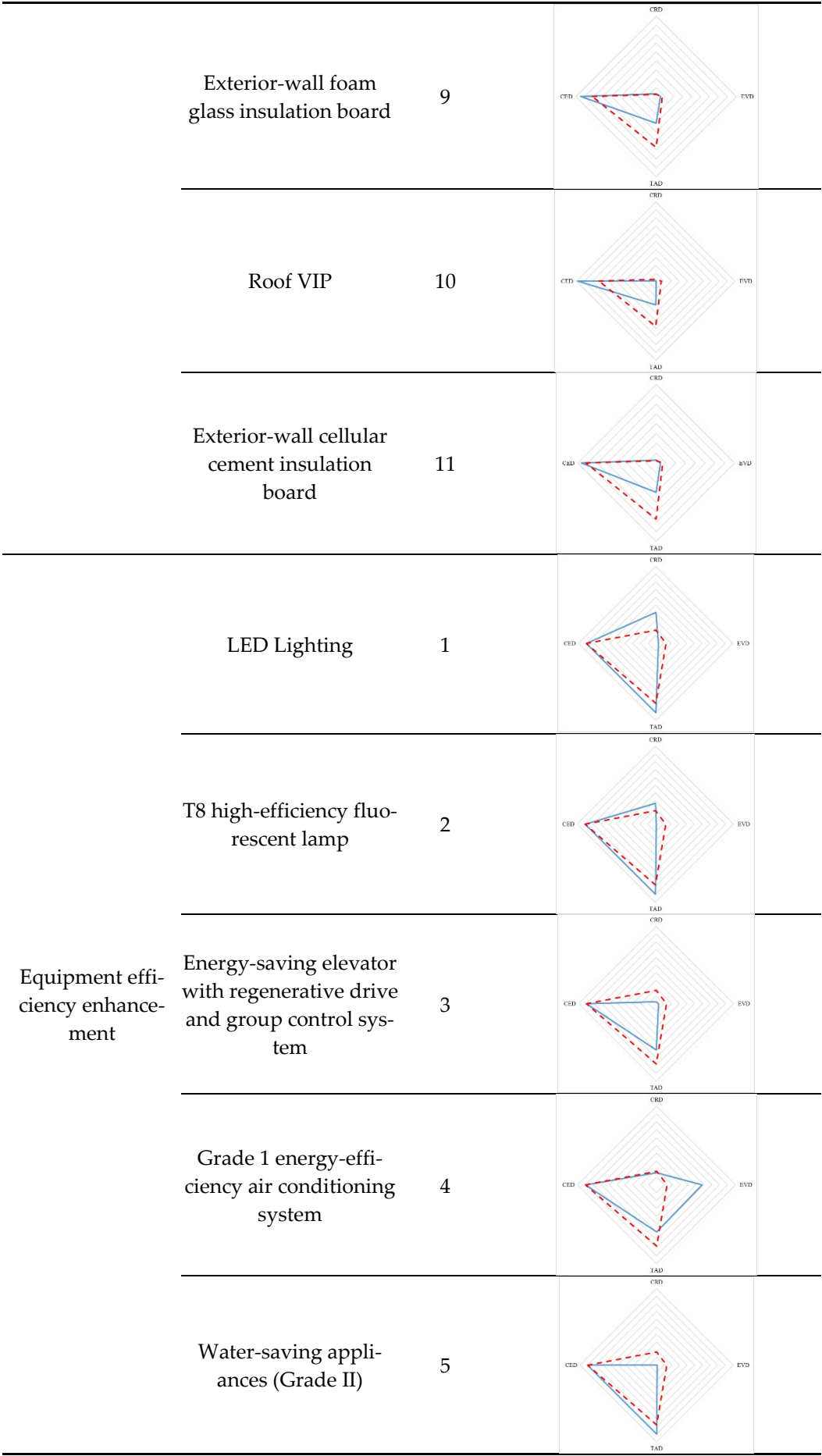
| Serial | Technology Name                                                                       | Comprehensive Score | Comprehensive Ranking |
|--------|---------------------------------------------------------------------------------------|---------------------|-----------------------|
| 17     | Rooftop PV system                                                                     | 0.80                | 1                     |
| 13     | LED lighting                                                                          | 0.62                | 2                     |
| 11     | Glazed section: thermal break aluminum frame with 6 + 12A + 6 + 0.15V + 6 Low—E glass | 0.59                | 3                     |
| 10     | Glazed section: 5mm single silver Low—E glass + 12A + 5 + 12A + 5                     | 0.58                | 4                     |
| 6      | Exterior-wall AAC blocks                                                              | 0.58                | 5                     |
| 7      | Roof XPS board                                                                        | 0.55                | 6                     |
| 1      | Exterior-wall rock wool insulation board                                              | 0.55                | 7                     |
| 14     | T8 high-efficiency fluorescent lamp                                                   | 0.54                | 8                     |
| 5      | Exterior-wall EPS board                                                               | 0.52                | 9                     |
| 15     | Energy-saving elevator with regenerative drive and group control system               | 0.49                | 10                    |
| 12     | Grade 1 energy-efficiency air conditioning system                                     | 0.49                | 11                    |
| 8      | Roof RPUF board                                                                       | 0.49                | 12                    |
| 18     | ASHP water heating system                                                             | 0.45                | 13                    |
| 2      | Exterior-wall glass wool insulation board                                             | 0.44                | 14                    |
| 4      | Exterior-wall foam glass insulation board                                             | 0.43                | 15                    |
| 9      | Roof VIP                                                                              | 0.41                | 16                    |
| 16     | Water-saving appliances (Grade II)                                                    | 0.40                | 17                    |
| 3      | Exterior-wall cellular cement insulation board                                        | 0.40                | 18                    |
| 19     | Rainwater recycling system                                                            | 0.21                | 19                    |

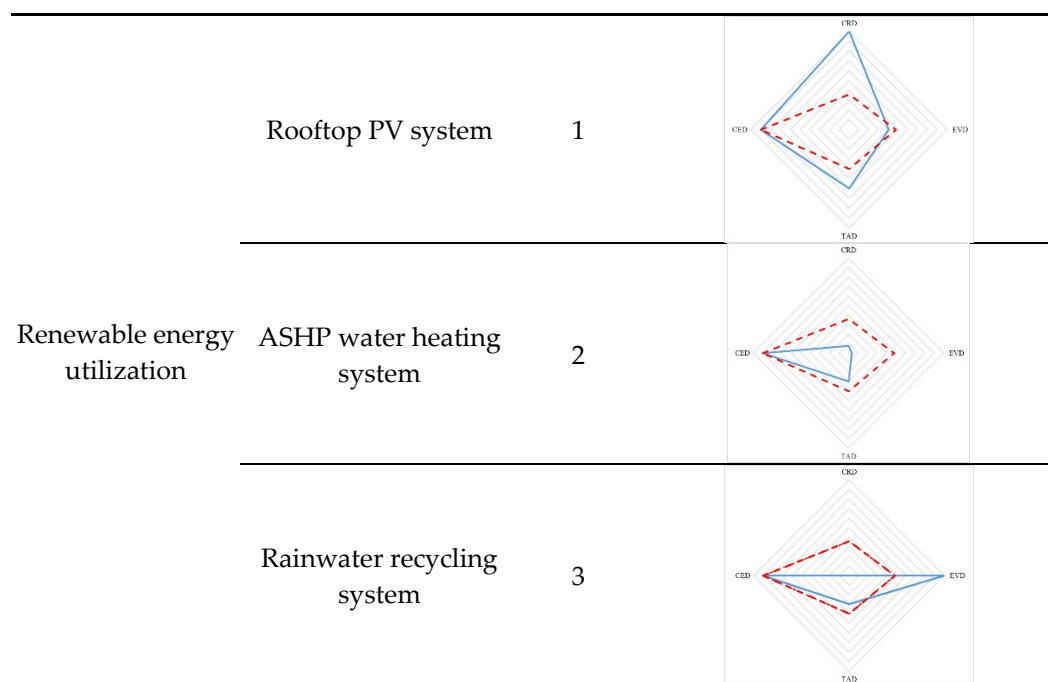
Notes: The green background indicates Envelope Structure Optimization technologies, the yellow background represents Equipment Efficiency Enhancement technologies, and the blue background indicates Renewable Energy Utilization technologies.

**Table 11.** Four-dimensional evaluation results of categorized technologies for Prototype 1.

| Technical Categories            | Technology Name                                                                         | Ranking | Four-Dimensional Evaluation Results                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------|
| Envelope structure optimization | Glazed section: thermal break aluminum frame with low-E glass (6 + 12A + 6 + 0.15V + 6) | 1       |  |







### 3.3.2. Analysis of Carbon Reduction Technologies for Mid-Rise Office Buildings

For a typical mid-rise office building (Prototype 2), based on the comprehensive ranking of low-carbon technologies (see Table 12) and the four-dimensional assessment results of the categorized technologies (see Table 13), the applicable carbon reduction technologies also fell into three categories: improving the thermal performance of the envelope structure, adopting high-efficiency equipment, and recycling and utilization of energy resources. Among these, the high-efficiency equipment category exhibits the best overall performance. Both LED lighting and T8 high-efficiency fluorescent lamps exhibit strong performance in terms of carbon reduction and applicability. The second category is energy and resource recycling and utilization. Similar to low-rise office buildings, rooftop PV systems have emerged as the optimal technical solution due to their high carbon reduction (energy conservation) performance. To improve the thermal performance of envelope structures, it is recommended to adopt roof extruded polystyrene boards and single-silver low-E glass (5 + 12A + 5 + 12A + 5). These two technologies show favorable performance in terms of carbon reduction and lower unit area carbon emissions.

Considering the three types of energy-saving technologies comprehensively, for mid-rise office buildings, the adoption of two high-efficiency lighting devices—LEDs and T8 high-efficiency fluorescent lamps—along with rooftop PV systems is the preferred solution for energy-saving renovations in similar office buildings. In terms of improving the thermal performance of the envelope structure, roof-extruded polystyrene boards should be prioritized.

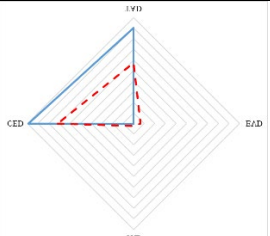
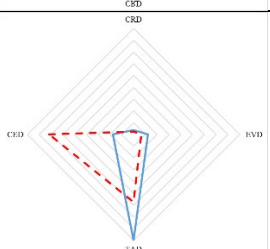
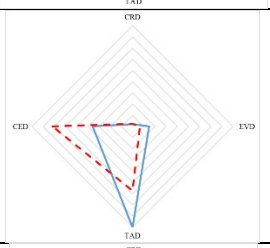
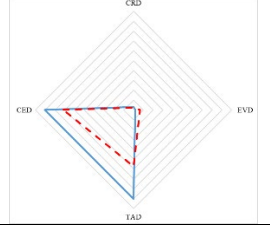
**Table 12.** Comprehensive score and ranking of low-carbon technologies for Prototype 2.

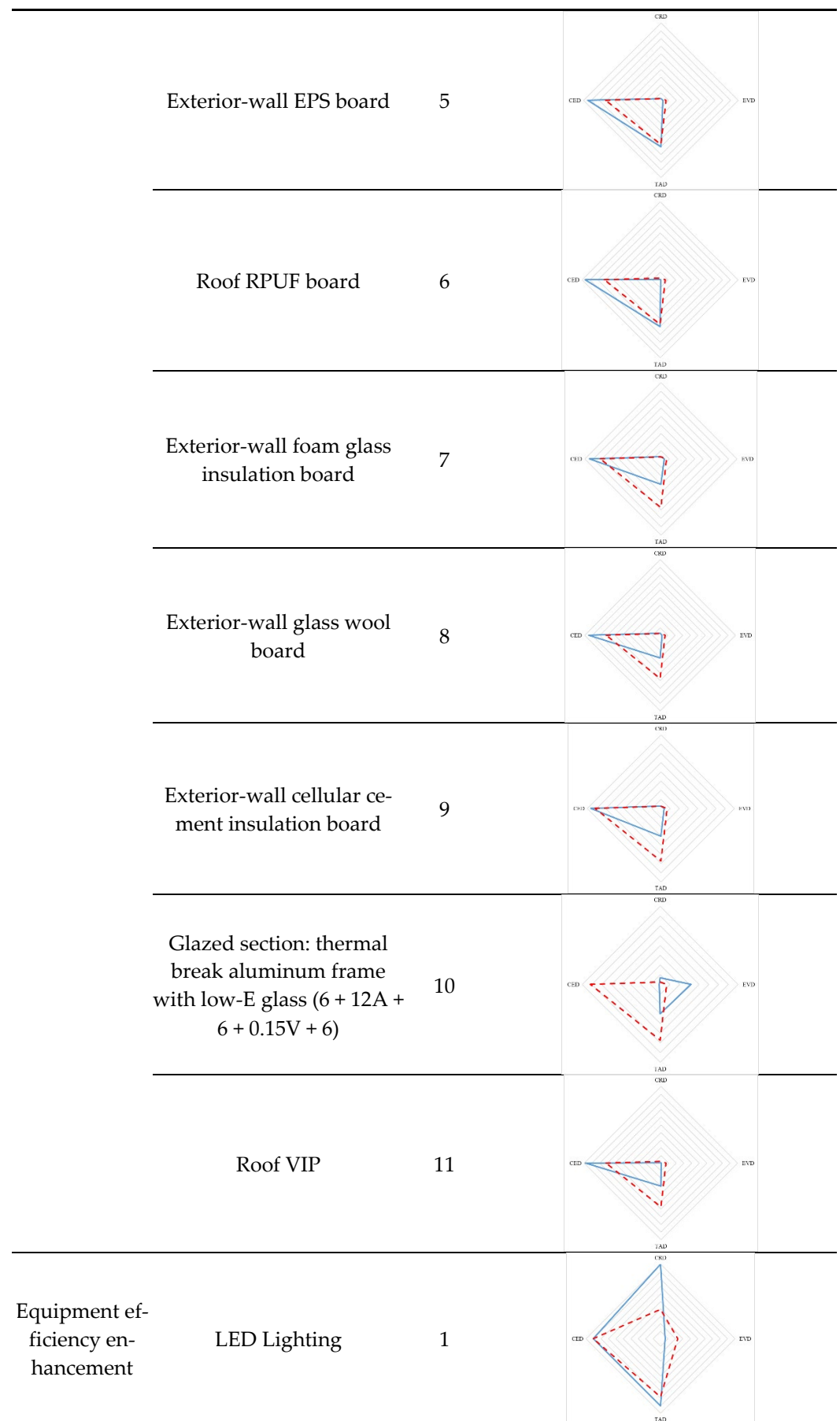
| Serial | Technology Name                                                   | Comprehensive Score | Comprehensive Ranking |
|--------|-------------------------------------------------------------------|---------------------|-----------------------|
| 13     | LED lighting                                                      | 0.87                | 1                     |
| 17     | Rooftop PV system                                                 | 0.70                | 2                     |
| 14     | T8 high-efficiency fluorescent lamp                               | 0.64                | 3                     |
| 7      | Roof XPS board                                                    | 0.62                | 4                     |
| 10     | Glazed section: single silver low-E glass (5 + 12A + 5 + 12A + 5) | 0.61                | 5                     |
| 6      | Exterior-wall AAC blocks                                          | 0.59                | 6                     |

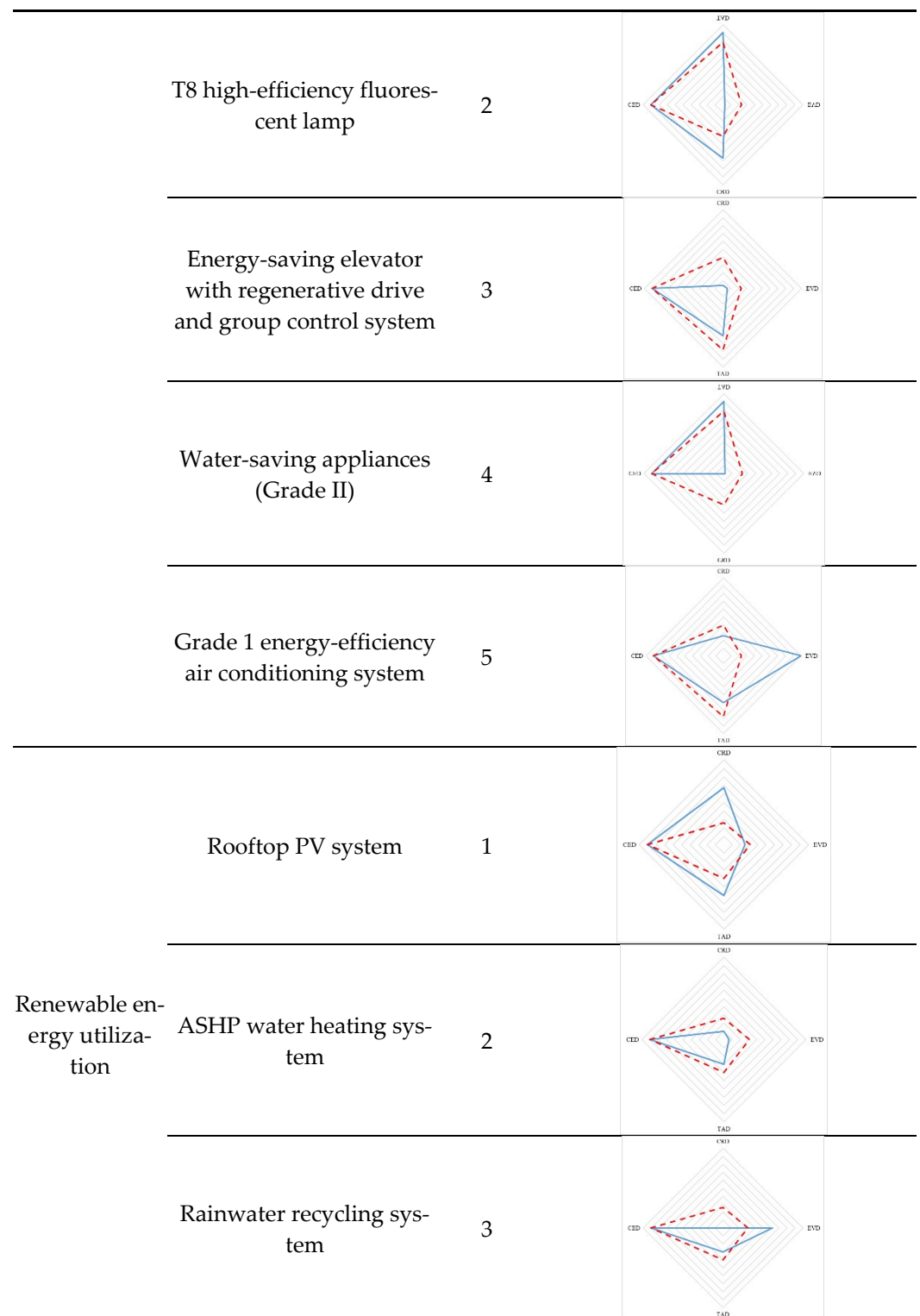
|    |                                                                                         |      |    |
|----|-----------------------------------------------------------------------------------------|------|----|
| 1  | Exterior-wall rock wool insulation board                                                | 0.55 | 7  |
| 15 | Energy-saving elevator with regenerative drive and group control system                 | 0.52 | 8  |
| 5  | Exterior-wall EPS board                                                                 | 0.51 | 9  |
| 16 | Water-saving appliances (Grade II)                                                      | 0.49 | 10 |
| 8  | Roof RPUF board                                                                         | 0.48 | 11 |
| 18 | ASHP water heating system                                                               | 0.47 | 12 |
| 4  | Exterior-wall foam glass insulation board                                               | 0.46 | 13 |
| 2  | Exterior-wall glass wool insulation board                                               | 0.45 | 14 |
| 3  | Exterior-wall cellular cement insulation board                                          | 0.45 | 15 |
| 11 | Glazed section: thermal break aluminum frame with low-E glass (6 + 12A + 6 + 0.15V + 6) | 0.41 | 16 |
| 9  | Roof VIP                                                                                | 0.40 | 17 |
| 12 | Grade 1 energy-efficiency air conditioning system                                       | 0.32 | 18 |
| 19 | Rainwater recycling system                                                              | 0.26 | 19 |

Notes: The green background indicates Envelope Structure Optimization technologies, the yellow background represents Equipment Efficiency Enhancement technologies, and the blue background indicates Renewable Energy Utilization technologies.

**Table 13.** Four-dimensional evaluation results of the categorized technologies for Prototype 2.

| Technical Categories            | Technology Name                                                   | Ranking | Four-Dimensional Evaluation Results                                                   |
|---------------------------------|-------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------|
| Envelope structure optimization | Roof XPS board                                                    | 1       |  |
|                                 | Glazed section: single silver low-E glass (5 + 12A + 5 + 12A + 5) | 2       |  |
|                                 | Exterior-wall AAC blocks                                          | 3       |  |
|                                 | Exterior-wall rock wool board                                     | 4       |  |





### 3.3.3. Analysis of Carbon Reduction Technologies for High-Rise Office Buildings

For a typical high-rise office building (Prototype 3), based on the comprehensive ranking of low-carbon technologies (see Table 14) and the four-dimensional assessment results of the categorized technologies (see Table 15), the applicable carbon reduction technologies are similar to those used in mid-rise structures. The high-efficiency equipment category exhibits the best overall performance. Both LED lighting and T8 high-efficiency fluorescent lamps exhibit strong performance in terms of carbon reduction and applicability. Rooftop PV power generation remains the optimal technical choice for energy and resource recycling and utilization due to its high carbon reduction (energy conservation)

performance. To improve the thermal performance of the building envelope, it is recommended to adopt single silver low-E glass (5 + 12A + 5 + 12A + 5) and install roof extruded polystyrene boards. These two technologies have become the preferred options for enhancing envelope thermal performance due to their excellent carbon reduction (energy conservation) capabilities and relatively favorable cost-effectiveness.

Considering the three categories of energy-saving technologies comprehensively, for high-rise office buildings, the adoption of two high-efficiency devices—LED lighting and T8 high-efficiency fluorescent lamps—along with rooftop PV systems, is the preferred solution for energy-saving renovations in similar office buildings. In terms of improving the envelope structure, single silver low-E glass (5 + 12A + 5 + 12A + 5 mm) is a priority recommendation.

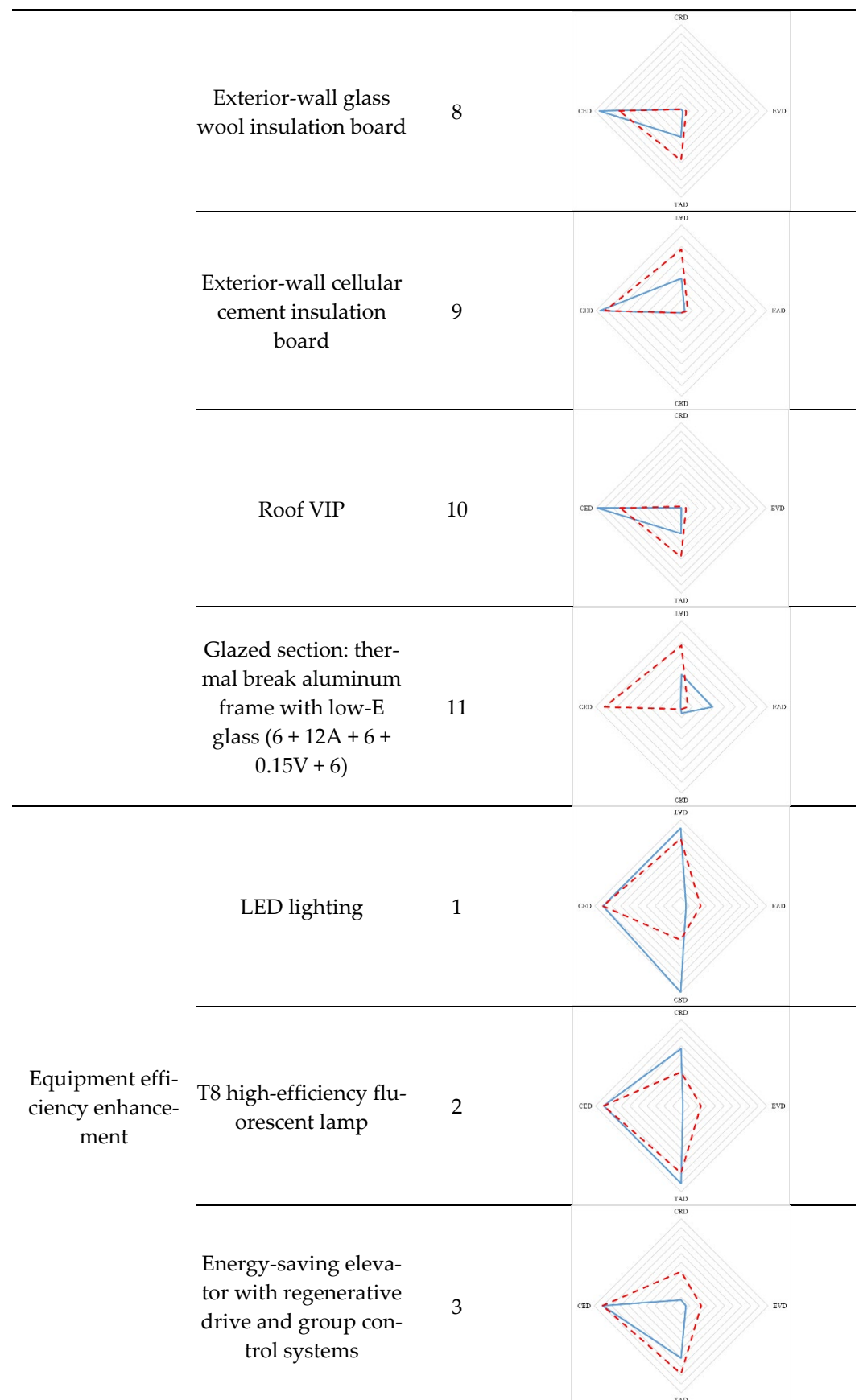
**Table 14.** Comprehensive score and ranking of low-carbon technologies for Prototype 3.

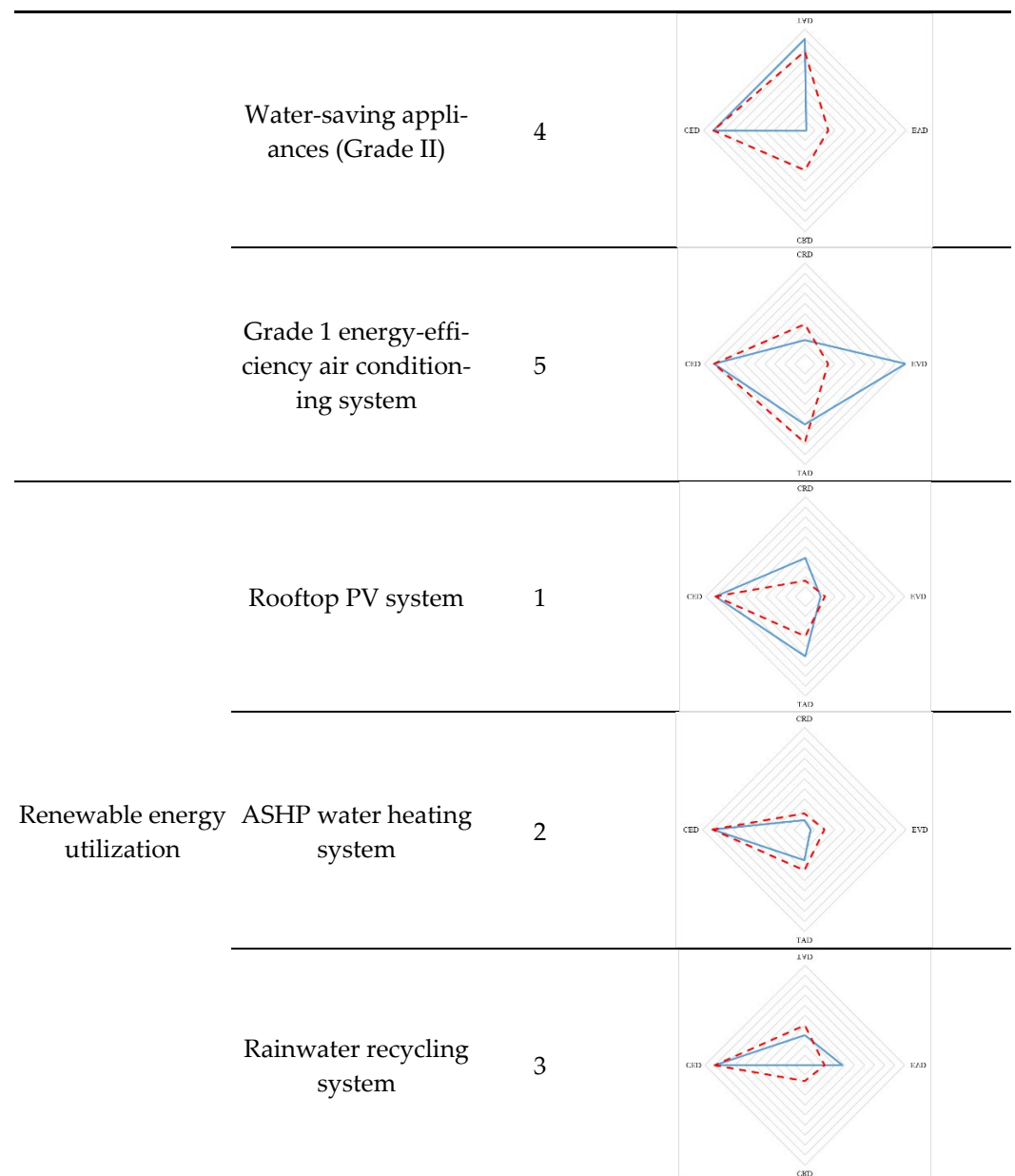
| Serial | Technology Name                                                                         | Comprehensive Score | Comprehensive Ranking |
|--------|-----------------------------------------------------------------------------------------|---------------------|-----------------------|
| 13     | LED lighting                                                                            | 0.88                | 1                     |
| 17     | Rooftop PV system                                                                       | 0.67                | 2                     |
| 14     | T8 high-efficiency fluorescent lamp                                                     | 0.65                | 3                     |
| 10     | Glazed section: single silver low-E glass (5 + 12A + 5 + 12A + 5)                       | 0.61                | 4                     |
| 7      | Roof XPS insulation board                                                               | 0.60                | 5                     |
| 6      | Exterior-wall AAC blocks                                                                | 0.58                | 6                     |
| 15     | Energy-saving elevator with regenerative drive and group control systems                | 0.56                | 7                     |
| 1      | Exterior-wall rock wool insulation board                                                | 0.56                | 8                     |
| 5      | Exterior-wall EPS board                                                                 | 0.52                | 9                     |
| 16     | Water-saving appliances (Grade II)                                                      | 0.50                | 10                    |
| 8      | Roof RPUF insulation board                                                              | 0.50                | 11                    |
| 18     | ASHP water heating system                                                               | 0.48                | 12                    |
| 4      | Exterior-wall Foam glass insulation board                                               | 0.47                | 13                    |
| 2      | Exterior-wall glass wool insulation board                                               | 0.46                | 14                    |
| 3      | Exterior-wall cellular cement insulation board                                          | 0.46                | 15                    |
| 9      | Roof VIP                                                                                | 0.44                | 16                    |
| 11     | Glazed section: thermal break aluminum frame with Low-E glass (6 + 12A + 6 + 0.15V + 6) | 0.42                | 17                    |
| 19     | Rainwater recycling system                                                              | 0.26                | 18                    |
| 12     | Grade 1 energy-efficiency air conditioning system                                       | 0.26                | 19                    |

Notes: The green background indicates Envelope Structure Optimization technologies, the yellow background represents Equipment Efficiency Enhancement technologies, and the blue background indicates Renewable Energy Utilization technologies.

**Table 15.** Four-dimensional evaluation results of categorized technologies for Prototype 3.

| Technical Categories            | Technology Name                                                   | Ranking | Four-Dimensional Evaluation Results |
|---------------------------------|-------------------------------------------------------------------|---------|-------------------------------------|
| Envelope structure optimization | Glazed section: single silver low-E glass (5 + 12A + 5 + 12A + 5) | 1       |                                     |
|                                 | Roof XPS board                                                    | 2       |                                     |
|                                 | Exterior-wall AAC blocks                                          | 3       |                                     |
|                                 | Exterior wall rock wool board                                     | 4       |                                     |
|                                 | Exterior-wall EPS board                                           | 5       |                                     |
|                                 | Roof RPUF board                                                   | 6       |                                     |
|                                 | Exterior-wall foam glass insulation board                         | 7       |                                     |





## 4. Discussions

### 4.1. Analysis of the Development Direction of Adaptive Technologies in Beijing

Using the AHP-Entropy Weight-TOPSIS hybrid model, the comprehensive scores and rankings of 19 technologies across three categories were calculated (see Table 16). Based on the analysis results and considering Beijing's cold climate characteristics (heating degree days HDD18 = 2890) and the green building development requirements outlined in the "14th Five-Year Plan", differentiated promotion strategies are proposed for adoption in the three technical clusters: optimizing the thermal performance of envelope structures, improving the energy efficiency of equipment systems, and promoting renewable energy utilization.

- (1) The promotion of energy efficiency improvement technologies for equipment systems is recommended to be prioritized to break the high carbon lock-in effect. Model calculations indicate that the average comprehensive score of equipment energy efficiency improvement technologies (0.485) is significantly higher than that of the other two types of technologies—passive technologies (0.321) and energy substitution technologies (0.364). Among these, lighting system renovation technologies,

such as LED lighting (0.625) and T8 fluorescent lamps (0.550), rank at the top. Their high adaptability can be attributed to several factors: outstanding carbon reduction benefits per unit investment (LED: 0.8 kgCO<sub>2</sub>/yuan), short renovation cycle (<3 months), and strong compatibility with BIM-based operation and maintenance systems.

- (2) The utilization of rooftop PV renewable energy is suggested for the active promotion of a resilient energy supply system. Among the renewable energy technologies, the rooftop PV system achieved the highest score (0.458). As a core component of distributed energy systems, the large-scale deployment of rooftop PV can significantly drive the comprehensive green transformation of economic and social development. It is estimated that the nationwide large-scale promotion of rooftop PV could achieve an average annual carbon reduction of over 1.4 billion tons, reduce electricity costs for urban users by more than 20%, and unlock over 1500 GW of demand potential for the PV industry. This initiative is expected to serve as a strategic breakthrough in overcoming the barriers to energy transition, promote green and inclusive energy consumption, and guide high-quality development in the PV industry.
- (3) Passive technological innovations and applications capable of precisely adapting to cold climates should be optimized and further enhanced. The design of building envelopes can reduce energy consumption and carbon emissions during building operation by lowering the building load demands. Although the comprehensive score of envelope structure technologies is relatively low (average: 0.321), their carbon reduction cost efficiency (0.18 kgCO<sub>2</sub>/yuan) is 3.2 times higher than that of equipment energy efficiency technologies. Among the various thermal performance optimization technologies for enclosure structures, mature external insulation solutions, such as rock wool insulation boards (0.397), roof XPS boards (0.395), and exterior wall glass wool boards (0.394)—demonstrate high potential but require strong policy support.

**Table 16.** Ranking and comprehensive scores for adaptive technology in Beijing and comparison with other methods.

| Ranking<br>(This<br>Study) | Technical Name                                                             | Integrating AHP-<br>Entropy-TOPSIS<br>Method (This<br>Study) | Comparison |             |
|----------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------|------------|-------------|
|                            |                                                                            |                                                              | TOPSIS     | AHP-Entropy |
| 1                          | LED Lighting                                                               | 0.625                                                        | 0.947      | 0.302       |
| 2                          | T8 high-efficiency fluorescent lamp                                        | 0.550                                                        | 0.839      | 0.261       |
| 3                          | Rooftop PV system                                                          | 0.458                                                        | 0.711      | 0.205       |
| 4                          | Exterior wall rock wool board                                              | 0.397                                                        | 0.632      | 0.161       |
| 5                          | Roof XPS board                                                             | 0.395                                                        | 0.635      | 0.154       |
| 6                          | Exterior wall glass wool board                                             | 0.394                                                        | 0.635      | 0.153       |
| 7                          | Water-saving appliances (Grade II)                                         | 0.385                                                        | 0.626      | 0.144       |
| 8                          | Exterior-wall EPS board                                                    | 0.357                                                        | 0.570      | 0.145       |
| 9                          | Roof RPUF board                                                            | 0.357                                                        | 0.570      | 0.143       |
| 10                         | Energy -saving elevator with re-generative drive and group control systems | 0.356                                                        | 0.567      | 0.145       |
| 11                         | Exterior-wall AAC block                                                    | 0.323                                                        | 0.531      | 0.115       |
| 12                         | Glazed section: single silver low-E glass (5 + 12A + 5 + 12A + 5)          | 0.310                                                        | 0.511      | 0.109       |
| 13                         | Roof VIP                                                                   | 0.308                                                        | 0.499      | 0.117       |

|    |                                                                                        |       |       |       |
|----|----------------------------------------------------------------------------------------|-------|-------|-------|
| 14 | ASHP water heating system                                                              | 0.306 | 0.494 | 0.118 |
| 15 | Exterior-wall cellular cement insulation board                                         | 0.293 | 0.465 | 0.121 |
| 16 | Exterior-wall foam glass insulation board                                              | 0.290 | 0.478 | 0.102 |
| 17 | Grade 1 energy-efficiency air conditioning system                                      | 0.266 | 0.445 | 0.088 |
| 18 | Rainwater recycling system                                                             | 0.220 | 0.368 | 0.073 |
| 19 | Glazed section: thermal break alumina frame with low-E glass (6 + 12A + 6 + 0.15V + 6) | 0.171 | 0.279 | 0.063 |

Notes: The green background indicates Envelope Structure Optimization technologies, the yellow background represents Equipment Efficiency Enhancement technologies, and the blue background indicates Renewable Energy Utilization technologies.

#### 4.2. Policy Recommendations for Beijing Low-Carbon Development

In recent years, China has introduced policies to promote green and low-carbon development in the construction industry. The “Opinions on Promoting Green Development in Urban and Rural Construction”, issued by the General Office of the State Council of China in 2021, clearly stipulates the implementation of carbon peaking and carbon neutrality actions within the construction sector. The “Work Plan for Accelerating Energy Conservation and Carbon Reduction in the Construction Sector”, co-issued by the National Development and Reform Commission of China (NDRC) and the Ministry of Housing and Urban-Rural Development of China (MOHURD) in 2024, further refines these targets by requiring that by 2030, all newly constructed office buildings must fully comply with green building standards and achieve an energy efficiency improvement of over 20% in existing buildings.

The four-dimensional evaluation framework directly aligns with China’s dual-carbon policy: CRD translates national absolute emission targets into project-level benchmarks; EVD quantifies abatement costs while balancing economic growth and decarbonization; TAD dynamically incorporates subsidies, grid factors, and technology learning rates to ensure roadmaps remain up to date; CID provides carbon intensity per unit floor area, offering a clear benchmark for sector-wide decarbonization strategies.

Based on the previous analysis of the development direction of regional adaptive technologies in Beijing, the promotion of carbon reduction technologies for office buildings in Beijing should focus on overcoming technological, economic, and institutional barriers through innovation in policy tools, thereby providing a model reference for the low-carbon transformation of buildings in megacities located in cold regions. Specifically, the following policy suggestions are proposed:

- (1) Suggestions for promoting energy efficiency improvement pathways: A digital carbon efficiency passport could be developed for equipment, integrating the full-lifecycle carbon trajectory (production—transportation—operation—recycling) and energy efficiency degradation curve. An Energy Efficiency Leader Program might then offer tiered subsidies for devices in the top 10% of the efficiency range (for every 5% above the benchmark value, increase subsidies by 15%). Complementarily, a digital twin platform for buildings could enable the real-time monitoring and mapping of equipment operation status and carbon emissions with a time resolution of  $\leq 15$  min.
- (2) Recommendations for large-scale adoption of rooftop PV distributed energy systems: Lightweight and flexible PV modules ( $\leq 3.5$  kg/m<sup>2</sup>) should be promoted to accommodate the load constraints of existing buildings. Support measures might include a

generation subsidy mechanism ( $\geq 0.35$  yuan/kWh), and a floor area ratio (FAR) reward mechanism for photovoltaic-integrated buildings (e.g., a 0.1 FAR reward per 100 kWh/m<sup>2</sup>·a of power generation). Block-level PV microgrids with a coverage radius of  $\leq 500$  m can facilitate the local consumption of surplus electricity (loss rate  $< 5\%$ ). It is also suggested to establish an intelligent microgrid system integrating consumption, production, storage, and regulation to support urban green and low-carbon renewal.

- (3) Promotion of carbon footprint access standards for external insulation materials and climate-responsive enclosure structures: Carbon footprint access standards (total lifecycle emissions  $\leq 18$  kgCO<sub>2</sub>/m<sup>2</sup>) can be established for external insulation materials. A thermal performance-linked FAR incentive mechanism may be introduced. For each 0.1 W/m<sup>2</sup>·K reduction in the heat transfer coefficient (U-value), a 0.05 FAR bonus is suggested.
- (4) Collaborative application of the three technology categories for specific office buildings: Given the synergies identified in Section 3.3, differentiated guidelines can be formulated for distinct office building typologies. Integrated design strategies, such as “passive design + intelligent systems” may then be selectively applied, providing flexible references for stakeholders, such as researchers, designers, engineers, and policymakers, to achieve targeted carbon reductions.

## 5. Conclusions

This study established a four-dimensional comprehensive evaluation framework for low-carbon technology in office buildings. Three typical office building models in the Beijing area were developed, and 19 applicable green and low-carbon technologies were selected. Pairwise comparative evaluations were conducted based on four key indicators: CRD, EVD, TAD, and CID. Furthermore, the evaluation values of each technology were categorized and systematically analyzed to determine their recommendation levels under the three typical building models across four-dimensional indicator systems. Finally, a combined method incorporating the AHP and entropy weight methods, together with an improved TOPSIS algorithm, was applied to comprehensively assess of these technologies. The main findings are as follows:

- (1) The research findings reveal distinct patterns in the selection of low-carbon technology solutions across different types of office buildings, providing customized combinations for three representative building categories. For low-rise office buildings, key technologies include roof PV systems, LED lighting, and thermal-break aluminum frames with low-E double-glazed laminated glass (6 + 12A + 6 + 0.15V + 6). In mid-rise and high-rise office buildings, LED lighting and T8 high-efficiency fluorescent lamps, along with rooftop PV systems, are prioritized. The primary difference lies in the envelope design: for mid-rise structures, enhanced roof insulation using XPS boards is emphasized; whereas, for high-rise buildings, double-glazed single silver low-E glass (5 + 12A + 5 + 12A + 5) is more strongly recommended.
- (2) Based on the final comprehensive scores of the three technology categories, LED lighting, T8 high-efficiency fluorescent lamps, and rooftop PV technology achieved consistently high rankings in both evaluation methods, demonstrating superior overall performance. Therefore, these technologies are prioritized for carbon reduction in office buildings in the Beijing region. The exterior wall rock wool board, roof XPS board, and exterior wall glass wool board also received relatively high scores within the building envelope system, indicating strong multidimensional performance, making them suitable as secondary recommended options.

- (3) Based on the analysis of regional adaptive technology development in Beijing, this study proposes strategies to improve energy efficiency and implement carbon monitoring, promote the large-scale adoption of rooftop PV systems, establish carbon footprint access standards for external insulation materials, and advance climate-responsive envelope structures. Policy recommendations, such as coordinated carbon reduction across the three technology categories, provide guidance for reducing carbon emissions in office buildings in Beijing.

This study innovatively develops a decision-making framework for carbon reduction technologies based on multidimensional evaluation and introduces a selection methodology for building carbon reduction solutions that integrate regional adaptability. This approach enables a more comprehensive and objective assessment of carbon reduction technology choices across different building types and regions, thereby providing both technical selection tools and a policy foundation for reducing carbon emissions in office buildings in Beijing and other similar urban areas. Furthermore, it outlines specific low-carbon development pathways tailored to distinct building types, establishing a replicable decision-support framework for the decarbonization of offices and other public buildings in megacities. The framework can be applied to the selection of low-carbon technologies for other types or regions of buildings, thereby accelerating the low-carbon transformation process in the global construction industry.

This study also has certain limitations. First, the method for selecting low-carbon renovation technologies developed in this research is based on a case study of Beijing. Its applicability is primarily limited to the cold climate zone and does not cover all the climate regions in China. Second, the low-carbon transformation of buildings involves a wide range of technologies. This study focuses only on key representative technologies and does not encompass all sub-technologies within the different technical categories. Future research should aim to enhance the adaptability of the proposed methods across diverse climate zones and expand their coverage to include a broader range of renovation technologies. This would enable the development of more precise low-carbon transformation strategies and contribute effectively to achieving China's major strategic goals of "carbon peak by 2030" and "carbon neutrality by 2060".

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